

# On Generalized Mersenne Primes in Real Quadratic Fields

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This Project was Submitted in Partial Fulfillment of the Requirements for  
the Bachelor's Degree of Science in Mathematics

School of Sciences / Mathematics Department  
The University of Jordan

June 7, 2026

# COMMITTEE DECISION

This Graduation Project (**On Generalized Mersenne Primes In Real Quadratic Fields**) was Successfully Defended and Approved on **June 7, 2026**.

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# Abstract

This project studies a generalization of classical Mersenne primes from the standard integers to the ring of integers  $\mathcal{O}_K$  of real quadratic fields  $K = \mathbb{Q}(\sqrt{d})$  where  $d$  is a square-free integer. Classical Mersenne numbers are defined as  $M_n = 2^n - 1$ , where  $n$  is a positive integer. When such a number is prime, it is called a Mersenne prime. This project studies analogues of Mersenne primes in quadratic fields. Specifically, we consider numbers of type  $M_{p,\alpha} = \frac{\alpha^p - 1}{\alpha - 1}$ , where  $\alpha \in \mathcal{O}_K$  is irreducible,  $\alpha - 1$  is a unit not equal to  $\pm 1$ , and  $p > 1$  is a rational prime. In this setting,  $M_{p,\alpha}$  is considered a generalized Mersenne prime when its absolute norm  $|N(M_{p,\alpha})|$  is a rational prime. Assuming  $\mathcal{O}_K$  is a principal ideal domain, this study investigates necessary conditions for the existence of such prime. We focus especially on the fields  $\mathbb{Q}(\sqrt{2})$  and  $\mathbb{Q}(\sqrt{3})$ . Finally, Mathematica is used to generate computational data that help identify and verify the norms of these generalized Mersenne primes for different prime exponents.

# Introduction

A positive integer is called a perfect number if it is equal to the sum of its proper positive divisors. Perfect numbers have a classical and important connection with Mersenne primes, this connection begins with Euclid's ancient formula and continuing with Euler's proof that every even perfect number arises from a Mersenne prime. Euler proved this result around 1750, although it was published nearly one hundred years later. In 1644, Marin Mersenne made a list of these special primes, and claimed that  $2^p - 1$  is prime for exactly eleven values of  $p \leq 257$ . However, since Mersenne and other early mathematicians had to do all their difficult calculations entirely by hand, his list contained several mistakes and also missed a few numbers. A major change occurred in 1952 when Raphael Robinson used an early computer to find five new primes in just one year, shifting the study of Mersenne primes from hand calculations toward computer-assisted methods. It is still unproven whether there are infinitely many Mersenne primes. Consequently, we currently only know of even perfect numbers, while the existence of odd perfect numbers remains an open question.

This study focuses on generalizing Mersenne primes in real quadratic fields  $K = \mathbb{Q}(\sqrt{d})$ , where  $d > 0$  is a square-free integer. It raises the question of how Mersenne primes can be defined and studied outside the ordinary integers  $\mathbb{Z}$ . In particular, the project studies the ring of integers  $\mathcal{O}_K$  in real quadratic fields under the assumption that  $\mathcal{O}_K$  is a principal ideal domain. Following the definition introduced in [7], we consider generalized Mersenne numbers of the form  $M_{p,\alpha} = \frac{\alpha^p - 1}{\alpha - 1}$  where  $\alpha \in \mathcal{O}_K$  such that  $\alpha$  is irreducible and  $\alpha - 1$  is a unit different from  $\pm 1$ , and  $p$  is a rational prime. Instead of using the classical base 2, this construction uses an element  $\alpha$  from the ring of integers  $\mathcal{O}_K$ . The main objective is to determine when  $M_{p,\alpha}$  has a rational prime absolute norm and therefore can be considered a generalized Mersenne prime. For more details, see [3].

This project is organized as follows:

Chapter 1 is divided into two sections : number theory and ring theory, each section introduces the basic definitions, theorems, and results that we needed in the next chapter.

Chapter 2 is divided into four sections:

Section 1 introduces classical Mersenne primes of the form  $M_n = 2^n - 1$ . It discusses their history, from Marin Mersenne's work to the computer era, and explains their well known connection with even perfect numbers.

Section 2 moves to quadratic number fields  $\mathbb{Q}(\sqrt{d})$ , explaining the difference between real and imaginary number fields. It also covers essential concepts like minimal polynomials, the ring of integers, conjugates, and how to find units and irreducible elements using the norm function.

Section 3 is devoted to Mersenne prime analogues in real quadratic fields. Assuming that  $\mathcal{O}_K$  is a principal ideal domain, we follow the definition introduced in [7] for the generalized Mersenne numbers  $M_{p,\alpha}$ . Since some statements in the article require clarification or additional assumptions, this section examines some main results carefully, corrects them where needed, and supports the discussion with detailed proofs and examples.

Section 4 presents Mathematica programs used to find irreducible elements in  $\mathcal{O}_K$  and the corresponding generalized Mersenne primes.

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# Chapter 1

## Preliminaries

### 1.1 Number Theory

In this section, we present the definitions and theorems required for the project, along with illustrative examples, for further details see [6] and [2].

**Definition 1.1.1** (Well-Ordering Principle). Every nonempty set  $S$  of nonnegative integers contains a least element; that is, there is some integer  $a$  in  $S$  such that  $a \leq b$  for all  $b$ 's belonging to  $S$ .

**Example 1.1.2.** Any random sample of positive integers has a least element.

**Theorem 1.1.1.** For any integer  $n$ ,  $n$  and  $n^2$  have the same parity. That is, if  $n$  is even, then  $n^2$  is even and if  $n$  is odd, then  $n^2$  is odd.

**Definition 1.1.3.** An integer  $b$  is said to be **divisible** by an integer  $a \neq 0$ , in symbols  $a \mid b$ , if there exists some integer  $c$  such that  $b = ac$ . We write  $a \nmid b$  to indicate that  $b$  is **not divisible** by  $a$ .

**Theorem 1.1.2.** For integers  $a, b, c$ , the following hold:

- (a)  $a \mid 0, 1 \mid a, a \mid a$ .
- (b)  $a \mid 1$  if and only if  $a = \pm 1$ .
- (c) If  $a \mid b$  and  $c \mid d$ , then  $ac \mid bd$ .
- (d) If  $a \mid b$  and  $b \mid c$ , then  $a \mid c$ .
- (e)  $a \mid b$  and  $b \mid a$  if and only if  $a = \pm b$ .
- (f) If  $a \mid b$  and  $b \neq 0$ , then  $|a| \leq |b|$ .
- (g) If  $a \mid b$  and  $a \mid c$ , then  $a \mid (bx + cy)$  for arbitrary integers  $x$  and  $y$ .

**Theorem 1.1.3** (Division Algorithm). For any integers  $a$  and  $b$ , with  $b > 0$ , there exist unique integers  $q$  and  $r$  satisfying

$$a = qb + r \quad 0 \leq r < b$$

The integers  $q$  and  $r$  are called, respectively, the quotient and remainder in the division of  $a$  by  $b$ .

**Definition 1.1.4.** Let  $a$  and  $b$  be two integers, with at least one of them different from zero. The **greatest common divisor** of  $a$  and  $b$ , denoted by  $\gcd(a, b)$ , is the positive integer  $d$  satisfying the following:

- (a)  $d \mid a$  and  $d \mid b$ .
- (b) If  $c \mid a$  and  $c \mid b$ , then  $c \leq d$ .

**Example 1.1.5.** Let  $a = 12$  and  $b = 18$ . The common divisors of 12 and 18 are  $\{1, 2, 3, 6\}$ . The greatest common divisor is  $d = 6$ , because:

- (a)  $6 \mid 12$  and  $6 \mid 18$ .
- (b) For any other common divisor  $c \in \{1, 2, 3\}$ , we have  $c < 6$ .

Thus,  $\gcd(12, 18) = 6$ .

**Theorem 1.1.4.** For any integer  $k \neq 0$ ,  $\gcd(ka, kb) = |k| \gcd(a, b)$ .

**Theorem 1.1.5.** Let  $r$  and  $m$  be any two nonzero integers, then  $\gcd(r, r + m) = \gcd(r, m)$ .

*Proof.* Let  $d_1 = \gcd(r, r + m)$  and let  $d_2 = \gcd(r, m)$ , want to show that  $d_1 \mid d_2$  and  $d_2 \mid d_1$  then  $d_1 = d_2$ .

Since  $d_1 = \gcd(r, r + m)$ , we have  $d_1 \mid r$  and  $d_1 \mid r + m$ . Hence  $d_1 \mid (r + m) - r$ . Thus  $d_1$  is a common divisor of  $r$  and  $m$ , so  $d_1 \mid d_2$ . Since  $d_2 = \gcd(r, m)$ , we have  $d_2 \mid r$  and  $d_2 \mid m$ , and hence  $d_2 \mid m + r$ . Since  $d_2$  is a common divisor of  $r$  and  $r + m$ ,  $d_2 \mid d_1$ . Therefore,  $d_1 = d_2$ .  $\square$

**Definition 1.1.6.** A positive integer  $n > 1$  is a **prime number** if its only positive factors are 1 and itself. A positive integer  $n > 1$  that is not a prime is a **composite number**.

**Theorem 1.1.6** (Fundamental Theorem of Arithmetic). Every positive integer  $n > 1$  is either a prime or a product of primes. This representation is unique up to the order of the factors.

**Example 1.1.7.** According to the Fundamental Theorem of Arithmetic, the integer  $n = 60$  has a prime factorization

$$60 = 2^2 \times 3 \times 5$$

Although the order of the factor may change, for example  $60 = 5 \cdot 2 \cdot 3 \cdot 2$ , the prime factors, including their multiplicity, are unique.

**Theorem 1.1.7.** There are infinitely many prime numbers.

*Proof.* Suppose that there are finitely many prime numbers. Let these primes be listed as:

$$K = \{p_1, p_2, p_3, \dots, p_n\}$$

Now, consider the integer  $t$  defined by:

$$t = (p_1 \cdot p_2 \cdot p_3 \cdots p_n) + 1$$

By Theorem 1.1.6,  $t$  must have at least one prime factor  $q$ .

If  $q$  is one of the primes in the list  $K$  (say  $p_i$ ), then  $q$  must divide the product  $(p_1 \cdot p_2 \cdots p_n)$ . Now, since  $q$  also divides  $t$ , then by Theorem 1.1.2 part (g),  $q$  must divide  $t - (p_1 \cdot p_2 \cdots p_n) = 1$ , a contradiction since there is no prime number that divides 1. Thus, the set of prime numbers must be infinite.  $\square$

**Theorem 1.1.8.** If  $a \mid bc$  with  $\gcd(a, b) = 1$ , then  $a \mid c$ .

**Example 1.1.8.** If  $6 \mid 5a$ , then  $6 \mid a$ , since  $\gcd(6, 5) = 1$ .

**Theorem 1.1.9.** If  $a \mid c$ ,  $b \mid c$ , and  $\gcd(a, b) = 1$ , then  $ab \mid c$ .

**Example 1.1.9.** If  $7 \mid a$  and  $11 \mid a$ , then  $77 \mid a$ .

**Theorem 1.1.10.** If  $\gcd(a, b) = 1$ , then  $\gcd(a^n, b^n) = 1$ , for all  $n \geq 1$ .

**Theorem 1.1.11.** Let  $n > 1$ , if  $2^n - 1$  is prime integer, then  $n$  is prime integer.

*Proof.* Assume that  $n$  is composite. This means we can write  $n$  as the product of two integers greater than 1:

$$n = a \cdot b \quad \text{where } 1 < a, b < n$$

Now, we substitute this into our expression:

$$2^n - 1 = 2^{ab} - 1 = (2^a)^b - 1$$

We can factor this expression using the standard algebraic identity for the difference of powers,  $x^b - 1 = (x - 1)(x^{b-1} + x^{b-2} + \cdots + x + 1)$ . Letting  $x = 2^a$ , we get:

$$2^n - 1 = (2^a - 1)(2^{a(b-1)} + 2^{a(b-2)} + \cdots + 2^a + 1)$$

Since  $a > 1$ , the first factor is greater than 1:

$$2^a - 1 > 2^1 - 1 = 1$$

Since  $b > 1$ , the second factor is also greater than 1:

$$2^{a(b-1)} + \cdots + 1 > 1$$

Because  $2^n - 1$  can be factored into two integers that are both greater than 1,  $2^n - 1$  must be composite, which is a contradiction.  $\square$

**Definition 1.1.10.** For a positive integer  $n$ , the function  $\sigma(n)$  denotes the **sum of all positive divisors of  $n$** .

**Definition 1.1.11.** A function  $f$  is said to be **multiplicative** if

$$f(mn) = f(m)f(n)$$

whenever  $\gcd(m, n) = 1$ .

**Theorem 1.1.12.** The function  $\sigma$  is multiplicative.

*Proof.* Let  $m$  and  $n$  be relatively prime integers. Because the result is trivially true if either  $m$  or  $n$  is equal to 1, we may assume that  $m > 1$  and  $n > 1$ . If

$$m = p_1^{k_1} p_2^{k_2} \cdots p_r^{k_r} \quad \text{and} \quad n = q_1^{j_1} q_2^{j_2} \cdots q_s^{j_s}$$

are the prime factorizations of  $m$  and  $n$ , then because  $\gcd(m, n) = 1$ , no  $p_i$  can occur among the  $q_j$ . It follows that the prime factorization of the product  $mn$  is given by

$$mn = p_1^{k_1} \cdots p_r^{k_r} q_1^{j_1} \cdots q_s^{j_s}$$

Applying Theorem 1.1.14, we get

$$\begin{aligned} \sigma(mn) &= \left[ \frac{p_1^{k_1+1} - 1}{p_1 - 1} \cdots \frac{p_r^{k_r+1} - 1}{p_r - 1} \right] \left[ \frac{q_1^{j_1+1} - 1}{q_1 - 1} \cdots \frac{q_s^{j_s+1} - 1}{q_s - 1} \right] \\ &= \sigma(m)\sigma(n) \end{aligned}$$

□

**Lemma 1.1.13.** Let  $p$  be a prime number and  $k \geq 1$  be an integer. Then:

$$\sigma(p^k) = 1 + p + p^2 + \cdots + p^k = \frac{p^{k+1} - 1}{p - 1}.$$

*Proof.* The only positive divisors of a prime power  $p^k$  are the powers of  $p$  up to  $k$ , namely

$$1, p, p^2, \dots, p^k.$$

Summing these divisors directly gives that:

$$\sigma(p^k) = 1 + p + p^2 + \cdots + p^k = \frac{p^{k+1} - 1}{p - 1}.$$

□

**Theorem 1.1.14.** If  $n = p_1^{k_1} p_2^{k_2} \cdots p_r^{k_r}$  is the prime factorization of  $n > 1$ , then

$$\sigma(n) = \frac{p_1^{k_1+1} - 1}{p_1 - 1} \frac{p_2^{k_2+1} - 1}{p_2 - 1} \cdots \frac{p_r^{k_r+1} - 1}{p_r - 1}.$$

**Example 1.1.12.** Consider the integer  $n = 180$ , which has the prime factorization:

$$180 = 2^2 \cdot 3^2 \cdot 5^1$$

The sum of the positive divisors of  $n$  is

$$\begin{aligned}\sigma(180) &= \left(\frac{2^3 - 1}{2 - 1}\right) \cdot \left(\frac{3^3 - 1}{3 - 1}\right) \cdot \left(\frac{5^2 - 1}{5 - 1}\right) \\ &= 7 \cdot 13 \cdot 6 = 546.\end{aligned}$$

**Definition 1.1.13.** A positive integer  $n$  is said to be **perfect** if  $n$  is equal to the sum of all its positive divisors, excluding  $n$  itself.

**Example 1.1.14.** Consider  $n = 28$ . The proper positive divisors of 28 are  $\{1, 2, 4, 7, 14\}$ . Then sum these divisors:

$$1 + 2 + 4 + 7 + 14 = 28$$

Since the sum of its proper divisors equals the number itself, 28 is a perfect number.

**Theorem 1.1.15.** A positive integer  $n$  is perfect if and only if  $\sigma(n) = 2n$ .

*Proof.* Let  $s(n)$  denote the sum of the proper positive divisors of  $n$ . Then

$$\sigma(n) = s(n) + n$$

Assume  $n$  is perfect. Then,  $s(n) = n$ . So

$$\sigma(n) = n + n = 2n.$$

Converely, assume  $\sigma(n) = 2n$ . Solving for  $s(n)$  gives:

$$s(n) = \sigma(n) - n = 2n - n = n.$$

Since the sum of the proper divisors of  $n$  equals  $n$ , it follows that  $n$  is perfect. □

**Definition 1.1.15.** An integer  $n \neq 1$  is said to be **square-free integer** if it is not divisible by the square of any integer greater than 1.

**Example 1.1.16.** The integer 10 is square-free, since no square greater than 1 divide it. On the other hand, 24 is not square-free because  $2^2 = 4$  divide 24.

**Theorem 1.1.16.** An integer  $n > 1$  is square-free if and only if  $n$  can be factored into a product of distinct primes.

**Example 1.1.17.** The integer  $n = 30$  is square-free because it can be factored into a product of distinct primes:  $30 = 2 \times 3 \times 5$ .

**Theorem 1.1.17.** An integer  $n > 1$  can be uniquely written as a product of a square free integer  $d$  and a perfect square  $k^2$ .

*Proof.* By Theorem 1.1.6,  $n$  has a unique prime factorization  $n = p_1^{a_1} \cdots p_t^{a_t}$ . For each  $i = 1, 2, \dots, t$ , write  $a_i = 2m_i + r_i$ , where  $r_i \in \{0, 1\}$ . Then

$$n = \left( \prod_{i=1}^t p_i^{r_i} \right) \cdot \left( \prod_{i=1}^t p_i^{m_i} \right)^2$$

Let  $d = \prod p_i^{r_i}$  and  $k = \prod p_i^{m_i}$  and since  $r_i$  is either 0 or 1, we get that  $d$  is square-free. The uniqueness of  $d$  and  $k$  follows directly from the uniqueness of the prime factorization of  $n$ .  $\square$

**Example 1.1.18.** The integer  $n = 72$ . Can be written as a product of a square-free integer and a perfect square. Indeed,  $72 = 2 \times 36$ , where 2 is square-free and  $36 = 6^2$  is a perfect square.

**Definition 1.1.19** (Modular arithmetic). Let  $m$  be a positive integer. Two integers  $a$  and  $b$  are said to be **congruent modulo  $m$**  if  $m \mid (a - b)$ . In this case, we write  $a \equiv b \pmod{m}$  and  $m$  is called the modulus of the congruence.

If  $m \nmid (b - a)$ , then  $a$  and  $b$  are said to be **incongruent modulo  $m$**  and we write  $a \not\equiv b \pmod{m}$ .

**Example 1.1.20.** Consider the modulus  $m = 6$ . We can investigate the congruence relation for different integers as follows:

1. **Congruent Case:** Let  $a = 20$  and  $b = 8$ . We find that  $a - b = 20 - 8 = 12$ . Since  $6 \mid 12$ , we say that 20 is congruent to 8 modulo 6, written as:

$$20 \equiv 8 \pmod{6}$$

2. **Incongruent Case:** Let  $a = 15$  and  $b = 5$ . We find that  $a - b = 15 - 5 = 10$ . Since  $6 \nmid 10$  (10 is not divisible by 6), we say that 15 is incongruent to 5 modulo 6, written as:

$$15 \not\equiv 5 \pmod{6}$$

**Theorem 1.1.18.** If  $\gcd(a, b) = 1$  and  $ab$  is a perfect square, then  $a$  and  $b$  are both perfect squares.

*Proof.* Write

$$a = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_r^{\alpha_r}, \quad b = q_1^{\beta_1} q_2^{\beta_2} \cdots q_s^{\beta_s},$$

where  $p_i$  and  $q_j$  are prime numbers. Since  $\gcd(a, b) = 1$ , we have

$$\{p_1, \dots, p_r\} \cap \{q_1, \dots, q_s\} = \emptyset.$$

Hence

$$ab = p_1^{\alpha_1} \cdots p_r^{\alpha_r} q_1^{\beta_1} \cdots q_s^{\beta_s}.$$

Because  $ab$  is a perfect square, every exponent in the prime factorization of  $ab$  is even. Therefore

$$\alpha_i \equiv 0 \pmod{2} \quad (1 \leq i \leq r), \quad \beta_j \equiv 0 \pmod{2} \quad (1 \leq j \leq s).$$

So there exist integers  $m_i, n_j \geq 0$  such that  $\alpha_i = 2m_i$  and  $\beta_j = 2n_j$ . Thus

$$a = (p_1^{m_1} p_2^{m_2} \cdots p_r^{m_r})^2, \quad b = (q_1^{n_1} q_2^{n_2} \cdots q_s^{n_s})^2.$$

Therefore,  $a$  and  $b$  are both perfect squares. □

**Theorem 1.1.19.** Let  $n > 1$  be fixed and  $a, b, c, d$  be arbitrary integers. Then the following properties hold:

- (a)  $a \equiv a \pmod{n}$ .
- (b) If  $a \equiv b \pmod{n}$ , then  $b \equiv a \pmod{n}$ .
- (c) If  $a \equiv b \pmod{n}$  and  $b \equiv c \pmod{n}$ , then  $a \equiv c \pmod{n}$ .
- (d) If  $a \equiv b \pmod{n}$  and  $c \equiv d \pmod{n}$ , then  $a + c \equiv b + d \pmod{n}$  and  $ac \equiv bd \pmod{n}$ .
- (e) If  $a \equiv b \pmod{n}$ , then  $a + c \equiv b + c \pmod{n}$  and  $ac \equiv bc \pmod{n}$ .
- (f) If  $a \equiv b \pmod{n}$ , then  $a^k \equiv b^k \pmod{n}$  for any positive integer  $k$ .

**Theorem 1.1.20.** If  $ca \equiv cb \pmod{n}$ , then  $a \equiv b \pmod{n/d}$ , where  $d = \gcd(c, n)$ .

**Example 1.1.21.** Consider the congruence  $12a \equiv 12b \pmod{30}$ . Since  $\gcd(12, 30) = 6$ , we get  $a \equiv b \pmod{30/6}$ , that is

$$a \equiv b \pmod{5}.$$

**Definition 1.1.22.** A **Diophantine equation** is a polynomial equation with integer coefficients in which the solutions when they exist are required to be integers.

**Definition 1.1.23.** An equation of the form  $ax + by = c$ , where  $a, b$ , and  $c$  are integers, is called a **linear Diophantine equation**.

**Definition 1.1.24** (Pell's Equation). **Pell's equation** is a Diophantine equation of the form

$$x^2 - Ny^2 = 1$$

where  $N$  is a positive non-square integer.

**Theorem 1.1.21.** Let  $N$  be a positive non-square integer. Then Pell's equation

$$x^2 - Ny^2 = 1$$

has infinitely many solutions in positive integers  $x$  and  $y$ .

## 1.2 Ring Theory

This section presents some basic definitions and theorems about rings, that will be needed for this project. For further details, see [5].

**Definition 1.2.1.** A **ring**  $R$  is a set with two binary operations, addition  $(a + b)$  and multiplication  $(ab)$ , satisfying the following axioms for all  $a, b, c \in R$ :

1.  $a + b = b + a$  (Commutativity of addition).
2.  $(a + b) + c = a + (b + c)$  (Associativity of addition).
3. There exists an additive identity  $0 \in R$  such that  $a + 0 = a$ .
4. For each  $a \in R$ , there exists an additive inverse  $-a \in R$  such that  $a + (-a) = 0$ .
5.  $a(bc) = (ab)c$  (Associativity of multiplication).
6.  $a(b + c) = ab + ac$  and  $(b + c)a = ba + ca$  (Distributive laws).

**Definition 1.2.2.** A subset  $S$  of a ring  $R$  is called a **subring** if  $S$  itself forms a ring under the same operations defined on  $R$ .

**Theorem 1.2.1** (Subring Test). Let  $R$  be a ring and  $S$  be a non-empty subset of  $R$ . Then  $S$  is a subring of  $R$  if the following two conditions hold for all elements  $a, b \in S$ :

1. **Closure under subtraction:**  
 $a - b \in S$
2. **Closure under multiplication:**  
 $ab \in S$

**Definition 1.2.3.** A ring  $R$  is called **commutative** if multiplication is commutative, meaning  $ab = ba$  for all  $a, b \in R$ .

**Definition 1.2.4.** A **unity** (or identity) in a ring is a nonzero element  $1 \in R$  such that  $1a = a1 = a$  for all  $a \in R$ .

**Example 1.2.5.** Let  $n \geq 2$  be an integer. The set of integers modulo  $n$ , denoted by:

$$\mathbb{Z}_n = \{0, 1, 2, \dots, n - 1\}$$

is a commutative ring with unity under addition and multiplication modulo  $n$ .

**Definition 1.2.6.** An element  $u$  in a ring with unity is called a **unit** if there exists  $v \in R$  such that  $uv = vu = 1$ .

**Definition 1.2.7.** In a commutative ring  $R$ , a nonzero element  $a$  is called a **zero-divisor** if there exists a nonzero element  $b \in R$  such that  $ab = 0$ .

**Example 1.2.8.** In the ring  $\mathbb{Z}_6$ , the elements 2 and 3 are zero-divisors because  $2 \neq 0$ ,  $3 \neq 0$ , and  $2 \cdot 3 = 0 \pmod{6}$ .

**Definition 1.2.9.** An **integral domain** is a commutative ring with unity that contains no zero-divisors.

**Definition 1.2.10.** A **field** is a commutative ring with unity where every nonzero element is a unit, that is every nonzero element has a multiplicative inverse.

**Example 1.2.11.** The set of complex numbers, denoted by  $\mathbb{C}$ , is defined as:

$$\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R} \text{ and } i^2 = -1\}$$

The set  $\mathbb{C}$  is a field under standard addition and multiplication.

**Definition 1.2.12.** Let  $F$  be a field. A non-empty subset  $K$  of  $F$  is called a **subfield** of  $F$  if  $K$  is itself a field under the same operations of addition and multiplication defined on  $F$ .

**Theorem 1.2.2** (Subfield Test). Let  $F$  be a field and  $K$  be a non-empty subset of  $F$ . Then  $K$  is a subfield of  $F$  if the following three conditions hold for all elements  $a, b \in K$ :

1. **Closure under subtraction:**

$$a - b \in K$$

2. **Closure under multiplication:**

$$ab \in K$$

3. **Closure under multiplicative inverse:**

$$a^{-1} \in K \text{ for every } a \neq 0$$

**Example 1.2.13.** Let  $d$  be a square-free integer with  $d \neq 1$ . Then

$$K = \mathbb{Q}(\sqrt{d}) = \{a + b\sqrt{d} \mid a, b \in \mathbb{Q}\}$$

is a field.

*Proof.* It is enough to show that  $K$  is a subfield of  $\mathbb{C}$ . Let  $\alpha, \beta \in K$  such that  $\alpha = a_1 + b_1\sqrt{d}$  and  $\beta = a_2 + b_2\sqrt{d}$ ,  $a_1, a_2, b_1, b_2 \in \mathbb{Q}$ .

First  $K$  is closed under subtraction, since

$$\alpha - \beta = (a_1 - a_2) + (b_1 - b_2)\sqrt{d} \in K$$

Also,  $K$  is closed under multiplication, since

$$\begin{aligned} \alpha\beta &= (a_1 + b_1\sqrt{d})(a_2 + b_2\sqrt{d}) \\ &= (a_1a_2 + b_1b_2d) + (a_1b_2 + a_2b_1)\sqrt{d} \in K \end{aligned}$$

Now, let  $r = a + b\sqrt{d} \neq 0$ . Then

$$r^{-1} = \frac{1}{a + b\sqrt{d}} = \frac{a - b\sqrt{d}}{(a + b\sqrt{d})(a - b\sqrt{d})} = \frac{a - b\sqrt{d}}{a^2 - b^2d}$$

We have  $a^2 - b^2d \neq 0$ . Otherwise  $d = (\frac{a}{b})^2$ , implying  $d$  is a perfect square in  $\mathbb{Q}$ , which contradicts that  $d$  is a square-free integer. Thus,  $r^{-1} \in K$ . Therefore,  $K$  is a field by Theorem 1.2.2. □

**Definition 1.2.14.** A subring  $A$  of a ring  $R$  is called an **ideal** of  $R$  if for every  $r \in R$  and  $a \in A$ , both  $ra$  and  $ar$  are in  $A$ .

**Theorem 1.2.3** (Ideal Test). A nonempty subset  $A$  of a ring  $R$  is an ideal of  $R$  if and only if:

1.  $a - b \in A$  for all  $a, b \in A$ .
2.  $ra \in A$  and  $ar \in A$  for all  $a \in A$  and  $r \in R$ .

**Definition 1.2.15.** Let  $R$  be a ring and  $A$  be an ideal of  $R$ . The set of all cosets  $R/A = \{r + A \mid r \in R\}$  forms a ring under coset addition and multiplication, called the **factor ring** (or quotient ring).

**Definition 1.2.16.** Let  $D$  be an integral domain. The **quotient field** of  $D$  is the field given by:

$$\left\{ \frac{a}{b} : a, b \in D, b \neq 0 \right\}$$

where the equivalence is defined as:

$$\frac{a}{b} = \frac{c}{d} \iff ad = bc$$

and the operations are:

$$\frac{a_1}{b_1} + \frac{a_2}{b_2} = \frac{a_1b_2 + a_2b_1}{b_1b_2}, \quad \frac{a_1}{b_1} \cdot \frac{a_2}{b_2} = \frac{a_1a_2}{b_1b_2}$$

**Definition 1.2.17.** A **ring homomorphism**  $\varphi : R \rightarrow S$  between two rings is a mapping that preserves both ring operations. That is, for all  $a, b \in R$ :

$$\varphi(a + b) = \varphi(a) + \varphi(b) \quad \text{and} \quad \varphi(ab) = \varphi(a)\varphi(b).$$

If  $\varphi$  is both injective and surjective, it is called a **ring isomorphism**.

**Theorem 1.2.4.** Let  $\varphi : R \rightarrow S$  be a ring isomorphism. Then  $R$  is a field if and only if  $S$  is a field.

**Definition 1.2.18.** Let  $R$  be a commutative ring with unity, and let  $a \in R$ . The ideal  $\langle a \rangle = \{ra \mid r \in R\}$  is called the **principal ideal generated by  $a$** .

**Definition 1.2.19.** A **prime ideal**  $A$  of a commutative ring  $R$  is a proper ideal ( $A \neq R$ ) such that if  $ab \in A$  for some  $a, b \in R$ , then  $a \in A$  or  $b \in A$ .

**Definition 1.2.20.** A **maximal ideal**  $A$  of a commutative ring  $R$  is a proper ideal such that if  $B$  is an ideal and  $A \subseteq B \subseteq R$ , then  $B = A$  or  $B = R$ .

**Theorem 1.2.5.** Let  $R$  be a commutative ring with unity. An ideal  $A$  of  $R$  is a prime ideal if and only if  $R/A$  is an integral domain.

**Theorem 1.2.6.** Let  $R$  be a commutative ring with unity. An ideal  $A$  of  $R$  is a maximal ideal if and only if  $R/A$  is a field.

Since every field is an integral domain, Theorem 1.2.5 and Theorem 1.2.6 imply the following result.

**Theorem 1.2.7.** In any commutative ring with unity, every maximal ideal is also a prime ideal.

**Definition 1.2.21.** Let  $D$  be an integral domain.

1. Two elements  $a, b \in D$  are **associates** if  $a = ub$  for some unit  $u \in D$ .
2. A nonzero, non-unit element  $p \in D$  is **prime** if  $p \mid bc$ , then  $p \mid b$  or  $p \mid c$ .
3. A nonzero, non-unit element  $a \in D$  is **irreducible** if it cannot be factored into two non-units, that is, if  $a = bc$ , then either  $b$  or  $c$  is a unit.

**Theorem 1.2.8.** In any integral domain, every prime element is irreducible.

*Proof.* Let  $D$  be an integral domain. Suppose that  $a$  is a prime element in  $D$ . To show that  $a$  is irreducible, assume  $a = bc$  for some  $b, c \in D$ . By the definition of a prime element, since  $a \mid bc$ , it follows that  $a \mid b$  or  $a \mid c$ . If  $a \mid b$ , there exists an element  $t \in D$  such that  $b = at$ , hence  $b = (bc)t = b(ct)$ . Since  $D$  is an integral domain and  $a$  is non-zero,  $b$  cannot be zero. By applying the cancellation law, we obtain  $1 = ct$ . This implies that  $c$  is a unit. Similarly, if  $a \mid c$ , it follows that  $b$  is a unit. Therefore,  $a$  is irreducible.  $\square$

**Definition 1.2.22.** A **principal ideal domain (PID)** is an integral domain in which every ideal is principal.

**Example 1.2.23.** The ring of integers  $\mathbb{Z}$  is a PID, as every ideal is of the form  $\langle n \rangle = n\mathbb{Z}$ .

**Definition 1.2.24.** Let  $R$  be a commutative ring with unity, and let  $a, b \in R$ . **The ideal generated by  $a$  and  $b$** , denoted by  $\langle a, b \rangle$ , is the set of all linear combinations:

$$\langle a, b \rangle = \{ax + by \mid x, y \in R\}$$

**Remark 1.2.1.** The ideal  $\langle a, b \rangle$  is the smallest ideal of  $R$  containing both  $a$  and  $b$ . That is, if  $I$  is any ideal of  $R$  such that  $a, b \in I$ , then  $\langle a, b \rangle \subseteq I$ .

**Theorem 1.2.9.** Let  $D$  be an integral domain, and let  $a, d \in D$ . The principal ideals generated by  $a$  and  $d$  are equal, that is  $\langle a \rangle = \langle d \rangle$  if and only if  $a$  and  $d$  are associates.

**Theorem 1.2.10.** In a principal ideal domain, an element is irreducible if and only if it is prime.

*Proof.* Let  $D$  be a principal ideal domain. Suppose that  $a$  is an irreducible element in  $D$ . To prove that  $a$  is prime, suppose  $a \mid bc$  for some  $b, c \in D$ . Consider the ideal  $I = \{ax + by \mid x, y \in D\}$ . Since  $D$  is a principal ideal domain, this ideal must be principal, meaning there exists some  $d \in D$  such that  $I = \langle d \rangle$ . Since  $a \in I$ , we can write  $a = dr$  for some  $r \in D$ . Because  $a$  is irreducible, either  $d$  is a unit or  $r$  is a unit. If  $d$  is a unit, then the ideal  $I$  is the entire ring  $D$ , which implies that  $1 \in I$ . Thus, we can write  $1 = ax + by$  for some  $x, y \in D$ . By multiplying both sides of this equation by  $c$  we get  $c = acx + bcy$ . Since  $a$  divides  $acx$  and  $a$  divides  $bcy$  by assumption,  $a$  must divide their sum, which means  $a \mid c$ . If  $r$  is a unit, then the ideals  $\langle a \rangle$  and  $\langle d \rangle$  are equal, so  $I = \langle a \rangle$ . Since  $b \in I$  by setting  $x = 0$  and  $y = 1$ , it follows that  $b \in \langle a \rangle$ . This means there exists an element  $t \in D$  such that  $b = at$ , showing that  $a \mid b$ . In either case,  $a \mid b$  or  $a \mid c$ , we get that  $a$  is a prime element. The converse is true by Theorem 1.2.8.  $\square$

**Definition 1.2.25.** An integral domain  $D$  is called a **unique factorization domain (UFD)** if:

1. Every nonzero, non-unit element of  $D$  can be written as a product of irreducibles.
2. This factorization is unique up to reordering factors and multiplication by units.

**Definition 1.2.26.** Let  $R$  be a commutative ring. The set of all polynomials with coefficients in  $R$ :

$$R[x] = \{a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 \mid a_i \in R, n \text{ is a non-negative integer}\}.$$

forms a ring under polynomial addition and multiplication, called the **ring of polynomials over  $R$** .

**Theorem 1.2.11.** If  $F$  is a field, then  $F[x]$  is a PID.

*Proof.* Let  $I$  be an ideal of  $F[x]$ . If  $I = \{0\}$ , then  $I = \langle 0 \rangle$ , which is principal. If  $I \setminus \{0\} \neq \emptyset$ , let  $A = \{\deg(p) \mid p \in I \setminus \{0\}\} \subseteq \mathbb{N} \cup \{0\}$ . By the Well-Ordering Principle,  $A$  has a least element. Let this least degree be  $\deg(f)$  for some  $f(x) \in I \setminus \{0\}$ . We claim that  $\langle f(x) \rangle = I$ .

Proof of claim: Since  $f(x) \in I$ , clearly  $\langle f(x) \rangle \subseteq I$ . Conversely, let  $g(x) \in I$ . By the Division Algorithm, there exist unique  $q(x), r(x) \in F[x]$  such that:  $g(x) = q(x)f(x) + r(x)$ , where  $r(x) = 0$  or  $\deg(r) < \deg(f)$ . So,  $r(x) = g(x) - q(x)f(x)$ . Since  $g(x) \in I$ ,  $f(x) \in I$ , and  $I$  is an ideal, we get  $r(x) \in I$ . If  $r(x) \neq 0$ , then

$\deg(r) < \deg(f)$ . However, this contradicts the fact that  $\deg(f)$  is the minimal degree in  $I \setminus \{0\}$ . Therefore,  $r(x)$  must be 0. If  $r(x) = 0$ , then  $g(x) = q(x)f(x)$ , which implies  $g(x) \in \langle f(x) \rangle$ . Thus,  $I \subseteq \langle f(x) \rangle$ . Therefore  $I = \langle f(x) \rangle$ , and hence  $F[x]$  is a PID.  $\square$

**Theorem 1.2.12.** Every principal ideal domain is a unique factorization domain.

**Example 1.2.27.** The ring of integers  $\mathbb{Z}$  is a classic example of a unique factorization domain. By the fundamental theorem of arithmetic, every integer other than 0, 1, and  $-1$  can be written as a product of prime numbers uniquely, up to the order of the factors and multiplication by units. For example,  $12 = 2 \cdot 2 \cdot 3$ . Other important examples of a UFDs are the polynomial rings  $\mathbb{Z}[x]$  and  $F[x]$ , where  $F$  is a field, such as  $\mathbb{R}[x]$  or  $\mathbb{Q}[x]$ .

**Theorem 1.2.13.** Let  $F$  be a field, and let  $f(x) \in F[x]$  be a polynomial of degree 2 or 3. Then  $f(x)$  is reducible over  $F$  if and only if  $f(x)$  has a root in  $F$ .

**Theorem 1.2.14 (Eisenstein's Criterion).** Let  $f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0 \in \mathbb{Z}[x]$ . If there exists a prime number  $p$  such that:  $p \nmid a_n$ , but  $p \mid a_i$  for all  $0 \leq i < n$ , and  $p^2 \nmid a_0$ , then  $f(x)$  is irreducible over the rational numbers  $\mathbb{Q}$ .

**Example 1.2.28.** The polynomial  $x^3 - 2 \in \mathbb{Z}[x]$  is irreducible over  $\mathbb{Q}$  by Eisenstein's Criterion, using the prime  $p = 2$ .

**Theorem 1.2.15.** Let  $D$  be an integral domain. Then  $U(D[x]) = U(D)$ .

**Theorem 1.2.16.** Let  $F$  be a field and let  $p(x) \in F[x]$ . The principal ideal  $\langle p(x) \rangle$  is a maximal ideal in  $F[x]$  if and only if  $p(x)$  is an irreducible polynomial over  $F$ .

*Proof.* Since  $F$  is a field, the polynomial ring  $F[x]$  is a principal ideal domain. First, assume that the principal ideal  $\langle p(x) \rangle$  is a maximal ideal in  $F[x]$ . By definition, a maximal ideal is a proper ideal, which implies that  $\langle p(x) \rangle \neq F[x]$ . Consequently,  $p(x)$  cannot be a non-zero constant since  $U(F[x]) = U(F)$ , so the degree of  $p(x)$  is at least one.

Now, suppose that  $p(x) = a(x)b(x)$  for some polynomials  $a(x), b(x) \in F[x]$ . This implies that  $p(x)$  belongs to the principal ideal  $\langle a(x) \rangle$ . Thus, we obtain  $\langle p(x) \rangle \subseteq \langle a(x) \rangle \subseteq F[x]$ . Because  $\langle p(x) \rangle$  is assumed to be a maximal ideal, we have the following cases.

Case 1: If  $\langle a(x) \rangle = \langle p(x) \rangle$ , then  $a(x)$  and  $p(x)$  are associates, so  $b(x)$  is a unit.

Case 2: If  $\langle a(x) \rangle = F[x]$ , then  $a(x)$  itself must be a unit.

In either case, any factorization of  $p(x)$  implies that one of the factors is a constant. Therefore,  $p(x)$  is irreducible over  $F$ .

Conversely, assume that  $p(x)$  is an irreducible polynomial over  $F$ . Since  $p(x)$  is irreducible, it is non-constant, and therefore the ideal  $\langle p(x) \rangle$  is a proper ideal of  $F[x]$ . Let  $I$  be an ideal of  $F[x]$  such that  $\langle p(x) \rangle \subseteq I \subseteq F[x]$ . Since  $F[x]$  is a principal ideal domain, every ideal in this ring is generated by a single element. Thus, there exists a polynomial  $g(x) \in F[x]$  such that  $I = \langle g(x) \rangle$ . Hence,  $\langle p(x) \rangle \subseteq$

$\langle g(x) \rangle \subseteq F[x]$ . We have  $\langle p(x) \rangle \subseteq \langle g(x) \rangle$  implies that  $g(x)$  divides  $p(x)$ . That is, there exists some polynomial  $q(x) \in F[x]$  such that  $p(x) = g(x)q(x)$ . Since  $p(x)$  is given to be irreducible over  $F$ , one of its factors must be a non zero constant. If  $g(x)$  is a constant, then it is a unit in  $F[x]$ , which means that its generated ideal  $I = \langle g(x) \rangle = F[x]$ . If  $q(x)$  is a constant, then  $g(x)$  and  $p(x)$  are associates, which means that  $I = \langle g(x) \rangle = \langle p(x) \rangle$ . Since any ideal  $I$  must be either  $\langle p(x) \rangle$  or  $F[x]$ , we conclude that  $\langle p(x) \rangle$  is a maximal ideal.  $\square$

# Chapter 2

## Generalized Mersenne Primes in Real Quadratic Fields

### 2.1 Mersenne Primes

Mersenne numbers are numbers of the form  $2^n - 1$  where  $n$  is in  $\mathbb{N}$ . When such a number is prime, it is called a Mersenne prime number.

These primes are named after Marin Mersenne, who studied them in 1644, and they are deeply connected to perfect numbers. Early mathematicians like Euclid and Euler proved that even perfect numbers are closely related to Mersenne primes. The discussion in this section is mainly based on [3].

**Definition 2.1.1.** A Mersenne number is a positive integer of the form

$$M_n = 2^n - 1$$

where  $n$  is in  $\mathbb{N}$ . If  $M_n$  is a rational prime, then it is called a Mersenne prime.

The following result illustrates the link between Mersenne primes and even perfect numbers.

**Theorem 2.1.1.** For an even positive integer  $n$ , the number  $n$  is perfect if and only if

$$n = 2^{p-1}(2^p - 1)$$

where  $2^p - 1$  is a rational prime.

*Proof.* Assume that  $2^p - 1$  is prime. Let

$$n = 2^{p-1}(2^p - 1).$$

Since  $\gcd(2^{p-1}, 2^p - 1) = 1$ , the divisor sum function is multiplicative, so

$$\sigma(n) = \sigma(2^{p-1})\sigma(2^p - 1).$$

By Lemma 1.1.13,

$$\sigma(2^{p-1}) = 1 + 2 + 2^2 + \cdots + 2^{p-1} = 2^p - 1.$$

Since  $2^p - 1$  is prime,

$$\sigma(2^p - 1) = 1 + (2^p - 1) = 2^p.$$

Hence,

$$\sigma(n) = (2^p - 1)(2^p) = 2 \cdot 2^{p-1}(2^p - 1) = 2n.$$

By Theorem 1.1.15, this implies that  $n$  is perfect. Therefore,

$$2^{p-1}(2^p - 1) \text{ is a perfect number.}$$

Conversely, let  $n$  be an even perfect number. Since  $n$  is even, we can factor out the highest power of 2 dividing  $n$ . Thus, we can express  $n$  as:

$$n = 2^{p-1}q$$

where  $p \geq 2$  and  $q$  is an odd integer such that  $\gcd(2^{p-1}, q) = 1$ .

Since  $n$  is a perfect number, by Theorem 1.1.15, we have  $\sigma(n) = 2n$ . Substituting our expression for  $n$  gives:

$$\sigma(2^{p-1}q) = 2(2^{p-1}q) = 2^p q$$

Because the divisor sum function  $\sigma$  is multiplicative and  $\gcd(2^{p-1}, q) = 1$ , we can separate the factors:

$$\sigma(2^{p-1})\sigma(q) = 2^p q$$

Using the result from Lemma 1.1.13 for the prime power  $2^{p-1}$ , we know that  $\sigma(2^{p-1}) = 2^p - 1$ . Substituting this into the equation gives:

$$(2^p - 1)\sigma(q) = 2^p q$$

Since  $\gcd(2^p, 2^p - 1) = 1$ , the term  $2^p - 1$  must divide the right-hand side of the equation. Because it cannot divide  $2^p$ , it must divide  $q$ . Therefore, there exists some integer  $d$  such that:

$$q = (2^p - 1)d$$

We now substitute  $q = (2^p - 1)d$  back into our main structural relation:

$$(2^p - 1)\sigma(q) = 2^p(2^p - 1)d$$

Dividing both sides by the common factor  $(2^p - 1)$ , we have:

$$\sigma(q) = 2^p d$$

Notice that we can rewrite the right-hand side of this equation as:

$$\sigma(q) = (2^p - 1)d + d = q + d$$

By definition,  $\sigma(q)$  is the sum of all positive divisors of  $q$ . We already know that  $q$  and  $d$  are two distinct positive divisors of  $q$ , since  $2^p - 1 > 1$  for  $p \geq 2$ , implying  $q > d$ .

Since  $\sigma(q) = q + d$ , it follows that  $q$  can't have positive divisors other than 1 and itself. This makes  $d = 1$ , which means that:

$$\sigma(q) = q + 1$$

Thus,  $q$  must be a prime number. Substituting  $d = 1$  back into the expression for  $q$ , we find:

$$q = 2^p - 1$$

Hence,  $2^p - 1$  is a prime number, and the even perfect number  $n$  is uniquely expressed in the form:

$$n = 2^{p-1}(2^p - 1)$$

□

The two directions of this theorem were established at different times by different mathematicians.

The forward direction,

$$2^p - 1 \text{ is rational prime implies } 2^{p-1}(2^p - 1) \text{ is perfect,}$$

This was known to the ancient Greeks and appears as Proposition 36 in Book IX of Euclid's Elements. This result Shows how to construct even perfect numbers from primes of the form  $2^p - 1$ .

The converse direction, state that every even perfect number has the form

$$2^{p-1}(2^p - 1) \text{ where } 2^p - 1 \text{ rational prime,}$$

This was proved by Euler around 1750. Euler showed that every even perfect number must arise from a Mersenne prime, thereby completing the characterization of even perfect numbers. Although this work was carried out around 1750, it was published nearly one hundred years later.

Table 2.1 illustrates that Mersenne correctly identified many exponents  $p \leq 257$  for which  $2^p - 1$  is prime. However, he made incorrect claims for  $p = 67$  and  $p = 257$ , where the corresponding Mersenne numbers are actually composite. Mersenne also missed to include the prime exponents  $p = 61, 89$ , and  $107$ , which were later shown to produce Mersenne primes.

| $p$ -exponent | Mersenne's claim<br>about $2^p - 1$ | Actual status<br>of $2^p - 1$ | Remark                    |
|---------------|-------------------------------------|-------------------------------|---------------------------|
| 2             | Prime                               | Prime                         | Correct                   |
| 3             | Prime                               | Prime                         | Correct                   |
| 5             | Prime                               | Prime                         | Correct                   |
| 7             | Prime                               | Prime                         | Correct                   |
| 13            | Prime                               | Prime                         | Correct                   |
| 17            | Prime                               | Prime                         | Correct                   |
| 19            | Prime                               | Prime                         | Correct                   |
| 31            | Prime                               | Prime                         | Correct                   |
| <b>61</b>     | <b>Composite</b>                    | <b>Prime</b>                  | <b>Missed by Mersenne</b> |
| <b>67</b>     | <b>Prime</b>                        | <b>Composite</b>              | <b>Incorrect claim</b>    |
| <b>89</b>     | <b>Composite</b>                    | <b>Prime</b>                  | <b>Missed by Mersenne</b> |
| <b>107</b>    | <b>Composite</b>                    | <b>Prime</b>                  | <b>Missed by Mersenne</b> |
| 127           | Prime                               | Prime                         | Correct                   |
| <b>257</b>    | <b>Prime</b>                        | <b>Composite</b>              | <b>Incorrect claim</b>    |

Table 2.1: Mersenne's Claims versus the actual primality of  $2^p - 1$  for  $p \leq 257$

This comparison shows both the importance and the limitations of Mersenne's early work. Even though he made some mistakes in his original list, his ideas inspired other mathematicians to continue the search for the Mersenne primes and helped develop a deeper understanding of their connection with perfect numbers.

#### Later Mathematician's Contribution in Mersenne Primes

1. Cataldi in 1588 verified that the 6-digit numbers, specifically  $2^{17} - 1$  and  $2^{19} - 1$  are both prime.
2. Euler in 1772 proved that the 10-digit number  $2^{31} - 1$  is prime.
3. Lucas in 1876, he introduced the largest Mersenne prime (at that time), namely  $2^{127} - 1$  is Mersenne prime. He proved its primality entirely by hand calculations.
4. Powers in 1914 proved that  $2^{107} - 1$  is prime.
5. Kraitchik in 1922 verified that  $2^{257} - 1$  is composite, but sadly his calculations were lost. Lehmer repeated the verification in 1927.
6. Robinson in 1952, he used the power of the computers to search for Mersenne primes. His program found five new Mersenne primes in that year, corresponding to the exponents 521, 607, 1279, 2203 and 2281.
7. Penk in 1979 found the first nontrivial factor of  $2^{257} - 1$ , namely 535006138114359.

For more details , see [3].

## 2.2 Quadratic Fields

Let  $d \neq 0, 1$  be a square-free integer. The quadratic field  $\mathbb{Q}(\sqrt{d})$  is defined by  $\mathbb{Q}(\sqrt{d}) = \{a + b\sqrt{d} : a, b \in \mathbb{Q}\}$ . These fields are called **real quadratic fields** if  $d > 0$ , and **imaginary quadratic fields** if  $d < 0$ . In this section, we study the structure of  $\mathbb{Q}(\sqrt{d})$  and its ring of integers  $\mathcal{O}_K$  and discuss some results related to units on irreducible elements in  $\mathcal{O}_K$ . For further details see [1] and [4]

**Definition 2.2.1.** Let  $d \neq 0, 1$  be a square-free integer. We define

$$K = \mathbb{Q}(\sqrt{d}) = \{a + b\sqrt{d} : a, b \in \mathbb{Q}\}.$$

**Lemma 2.2.1.** Let  $d \neq 0, 1$  be a square-free integer. Then the polynomial  $p(x) = x^2 - d \in \mathbb{Q}[x]$  is irreducible over  $\mathbb{Q}$ .

*Proof.* Suppose  $p(x)$  is reducible over  $\mathbb{Q}$ . Since  $\deg(p) = 2$ , this implies that  $p(x)$  has a root  $x_0 \in \mathbb{Q}$ . Hence  $p(x_0) = 0$ . So,  $x_0^2 - d = 0$  and thus  $x_0^2 = d$ , a contradiction since,  $d$  is a square-free integer. Therefore,  $p(x)$  has no roots in  $\mathbb{Q}$  and it is irreducible.  $\square$

**Lemma 2.2.2.** Let  $d \neq 0, 1$  be a square-free integer, and let  $p(x) = x^2 - d \in \mathbb{Q}[x]$ . Then the quotient ring  $\mathbb{Q}[x]/\langle p(x) \rangle$  is a field.

*Proof.* Since  $\mathbb{Q}$  is a field,  $\mathbb{Q}[x]$  is a PID. By Lemma 2.2.1  $p(x)$  is irreducible over  $\mathbb{Q}$ . Therefore, the ideal  $\langle p(x) \rangle$  is maximal ideal in  $\mathbb{Q}[x]$ . Therefore, by Theorem 1.2.6  $\mathbb{Q}[x]/\langle p(x) \rangle$  is a field.  $\square$

**Remark 2.2.1.** Every element of the quotient ring  $\mathbb{Q}[x]/\langle x^2 - d \rangle$  can be represented in the form  $[a + bx]$ , where  $[a + bx]$  is used as a short notation to denote the coset  $a + bx + \langle x^2 - d \rangle$ .

**Lemma 2.2.3.** The map  $\varphi : \mathbb{Q}[x]/\langle x^2 - d \rangle \rightarrow \mathbb{Q}(\sqrt{d})$  defined by  $\varphi([a + bx]) = a + b\sqrt{d}$  is a field isomorphism.

*Proof.* Suppose  $[A] = [B]$  in  $\mathbb{Q}[x]/\langle x^2 - d \rangle$ , where  $A = \alpha_1 + \beta_1 x$  and  $B = \alpha_2 + \beta_2 x$ . Then  $A - B \in \langle x^2 - d \rangle$  and hence  $(\alpha_1 - \alpha_2) + (\beta_1 - \beta_2)x = q(x)(x^2 - d)$  for some  $q(x) \in \mathbb{Q}[x]$ . Since the degree of the left-hand side is at most 1 when it is nonzero and the degree of  $(x^2 - d)$  is 2, we must have  $q(x) = 0$ . So  $(\alpha_1 - \alpha_2) + (\beta_1 - \beta_2)x = 0$ . So,  $\alpha_1 = \alpha_2$  and  $\beta_1 = \beta_2$ . Thus,  $\varphi([A]) = \varphi([B])$  and  $\varphi$  is well-defined. Now, let  $A = \alpha_1 + \beta_1 x$  and  $B = \alpha_2 + \beta_2 x$ . Then

$$\begin{aligned} \varphi([A] + [B]) &= \varphi([\alpha_1 + \alpha_2 + (\beta_1 + \beta_2)x]) \\ &= (\alpha_1 + \alpha_2) + (\beta_1 + \beta_2)\sqrt{d} \\ &= (\alpha_1 + \beta_1\sqrt{d}) + (\alpha_2 + \beta_2\sqrt{d}) \\ &= \varphi([A]) + \varphi([B]) \end{aligned}$$

Also,  $[A] \cdot [B] = [(\alpha_1\alpha_2 + \beta_1\beta_2d) + (\alpha_1\beta_2 + \alpha_2\beta_1)x]$ . So,  $\varphi([A][B]) = (\alpha_1\alpha_2 + \beta_1\beta_2d) + (\alpha_1\beta_2 + \alpha_2\beta_1)\sqrt{d}$ . Therefore,  $(\alpha_1 + \beta_1\sqrt{d})(\alpha_2 + \beta_2\sqrt{d}) = \varphi(A)\varphi(B)$ . Thus  $\varphi$  is a homomorphism. To show that  $\varphi$  is injective, suppose  $\varphi([A]) = \varphi([B])$ . Then  $\alpha_1 + \beta_1\sqrt{d} = \alpha_2 + \beta_2\sqrt{d}$ . And hence  $(\alpha_1 - \alpha_2) + (\beta_1 - \beta_2)\sqrt{d} = 0$ . Since  $d$  is square-free,  $\sqrt{d}$  is irrational. For this sum to be zero with rational coefficients, we must have  $\alpha_1 = \alpha_2$  and  $\beta_1 = \beta_2$ . Thus  $[A] = [B]$ . Finally let us show that  $\varphi$  is surjective. If  $y \in \mathbb{Q}(\sqrt{d})$ , then  $y = q + r\sqrt{d}$  for some  $q, r \in \mathbb{Q}$  and hence  $\varphi(q + rx) = q + r\sqrt{d}$  where  $q + rx \in \mathbb{Q}[x]$ . Thus,  $\varphi$  is a ring isomorphism.  $\square$

Although  $K = \mathbb{Q}(\sqrt{d})$  was shown to be a field in Example 1.2.13 by direct verification, we give here another proof using quotient ring.

**Theorem 2.2.4.** Let  $d \neq 0, 1$  be a square-free. Then  $K = \mathbb{Q}(\sqrt{d})$  is a field.

*Proof.* By Lemma 2.2.3, the quotient ring  $\mathbb{Q}[x]/\langle x^2 - d \rangle$  is a field. By Lemma 2.2.3,  $\mathbb{Q}[x]/\langle x^2 - d \rangle \cong K = \mathbb{Q}(\sqrt{d})$ . Therefore,  $K$  is a field by Theorem 1.2.4.  $\square$

The following theorem gives another way to describe  $K = \mathbb{Q}(\sqrt{d})$ . It shows that  $K$  is the field of fractions of  $\mathbb{Z}[\sqrt{d}]$ , and therefore gives another proof that  $K$  is a field.

**Definition 2.2.2.** Let  $d \neq 0, 1$  be a square-free integer. We define the ring  $\mathbb{Z}[\sqrt{d}]$  as:

$$\mathbb{Z}[\sqrt{d}] = \{m + n\sqrt{d} : m, n \in \mathbb{Z}\}.$$

**Remark 2.2.2.**  $\mathbb{Z}[\sqrt{d}]$  forms an integral domain under standard addition and multiplication.

**Theorem 2.2.5.** Let  $d \neq 0, 1$  be a square-free. Then  $\mathbb{Q}(\sqrt{d})$  is the quotient field of  $\mathbb{Z}[\sqrt{d}]$ . In particular, since

$$\mathbb{Q}(\sqrt{d}) = \left\{ \frac{a + b\sqrt{d}}{m + n\sqrt{d}} : a + b\sqrt{d}, m + n\sqrt{d} \in \mathbb{Z}[\sqrt{d}], m + n\sqrt{d} \neq 0 \right\}$$

where

$$\frac{a + b\sqrt{d}}{m + n\sqrt{d}} \cdot \frac{m - n\sqrt{d}}{m - n\sqrt{d}} = r + s\sqrt{d} \in \mathbb{Q}(\sqrt{d})$$

This quotient field is the smallest field containing  $\mathbb{Z}[\sqrt{d}]$ .

**Definition 2.2.3** (Monic Polynomial). A polynomial  $p(x) \in F[x]$  of degree  $n \geq 0$  is called **monic** if its leading coefficient is equal to 1. That is,  $p(x)$  is of the form:

$$p(x) = x^n + \sum_{i=0}^{n-1} a_i x^i = x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0$$

where  $a_i \in F$  for all  $i$ .

**Definition 2.2.4.** Let  $K$  be a subfield of  $\mathbb{C}$ . An element  $\alpha \in K$  is **integral over  $\mathbb{Z}$**  if there exists a monic polynomial

$$f(x) = x^n + c_{n-1}x^{n-1} + \dots + c_1x + c_0 \in \mathbb{Z}[x]$$

such that  $f(\alpha) = 0$ .

**Lemma 2.2.6.** If  $f(x) \in \mathbb{Q}[x]$ , then  $f(x)$  can be written in the form:

$$f(x) = \frac{u}{m}(a_nx^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0)$$

where  $a_i \in \mathbb{Z}$  share no common divisor greater than 1, that is  $\gcd(a_n, a_{n-1}, \dots, a_0) = 1$ , and  $\gcd(u, m) = 1$ .

*Proof.* Let  $f(x) \in \mathbb{Q}[x]$  be a polynomial of degree  $n$ :

$$f(x) = c_nx^n + c_{n-1}x^{n-1} + \dots + c_1x + c_0$$

Since each coefficient  $c_i$  is a rational number, we can write  $c_i = \frac{k_i}{m_i}$  where  $k_i, m_i \in \mathbb{Z}$ . Thus:

$$f(x) = \frac{k_n}{m_n}x^n + \frac{k_{n-1}}{m_{n-1}}x^{n-1} + \dots + \frac{k_0}{m_0}$$

By finding a common denominator  $M = m_nm_{n-1}\dots m_0$ , we can rewrite the polynomial as:

$$f(x) = \frac{1}{M}(a'_nx^n + a'_{n-1}x^{n-1} + \dots + a'_0)$$

where each  $a'_i \in \mathbb{Z}$ . Let  $v$  be the greatest common divisor of the integer coefficients:

$$v = \gcd(a'_n, a'_{n-1}, \dots, a'_0)$$

Factoring out  $v$ , we get:

$$f(x) = \frac{v}{M}(a_nx^n + a_{n-1}x^{n-1} + \dots + a_0)$$

where  $\gcd(a_n, a_{n-1}, \dots, a_0) = 1$ . Finally, by canceling any common factors between  $v$  and  $M$ , we obtain the simplified fraction  $\frac{u}{m}$  such that  $\gcd(u, m) = 1$ :

$$f(x) = \frac{u}{m}(a_nx^n + a_{n-1}x^{n-1} + \dots + a_0)$$

□

**Lemma 2.2.7.** The map  $\varphi : \mathbb{Z}[x] \rightarrow \mathbb{Z}_p[x]$  defined by  $\varphi(f(x)) = \overline{f(x)}$ , where  $\overline{f(x)}$  denotes the polynomial obtained by reducing the coefficients of  $f(x)$  modulo  $p$ , is a ring homomorphism.

*Proof.* Let  $f(x), g(x) \in \mathbb{Z}[x]$ . To show that  $\varphi$  is a ring homomorphism, we verify the following two properties:

1. **Additivity:** Since reduction modulo  $p$  is a homomorphism from  $\mathbb{Z} \rightarrow \mathbb{Z}_p$ , the coefficients of the sum are preserved:

$$\begin{aligned}\varphi(f(x) + g(x)) &= \overline{f(x) + g(x)} \\ &= \overline{f(x)} + \overline{g(x)} \\ &= \varphi(f(x)) + \varphi(g(x))\end{aligned}$$

2. **Multiplicativity:** Similarly, because the product of polynomials is defined by the sum and product of coefficients, and reduction modulo  $p$  preserves these operations:

$$\begin{aligned}\varphi(f(x) \cdot g(x)) &= \overline{f(x) \cdot g(x)} \\ &= \overline{f(x)} \cdot \overline{g(x)} \\ &= \varphi(f(x)) \cdot \varphi(g(x))\end{aligned}$$

Since both properties hold,  $\varphi$  is a ring homomorphism.  $\square$

**Lemma 2.2.8** (Gauss). If a monic polynomial  $f(x) \in \mathbb{Z}[x]$  factors into a product of two polynomials of lower degrees  $m$  and  $n$  in  $\mathbb{Q}[x]$ , then  $f(x)$  has such a factorization with polynomials of the same degrees  $m$  and  $n$  in  $\mathbb{Z}[x]$ .

*Proof.* Suppose  $f(x) = a(x)b(x)$  where  $a(x), b(x) \in \mathbb{Q}[x]$ . By applying Lemma 2.2.6 to both  $a(x)$  and  $b(x)$ , we can write:

$$a(x) = \frac{u_1}{m_1}(a_r x^r + \cdots + a_0), \quad b(x) = \frac{u_2}{m_2}(b_s x^s + \cdots + b_0)$$

where  $\gcd(u_i, m_i) = 1$  and  $\gcd(a_r, \dots, a_0) = \gcd(b_s, \dots, b_0) = 1$ . Let  $\frac{v}{w} = \frac{u_1 u_2}{m_1 m_2}$  in simplest form (such that  $\gcd(v, w) = 1$ ). Then:

$$f(x) = \frac{v}{w}(a_r x^r + \cdots + a_0)(b_s x^s + \cdots + b_0)$$

This implies:

$$wf(x) = v(a_r x^r + \cdots + a_0)(b_s x^s + \cdots + b_0) \tag{1}$$

**Case 1** If  $w = 1$ , then since  $f(x)$  is monic, we must have  $v = a_r b_s = 1$  or  $v = a_r b_s = -1$ . In the two cases,  $a_1(x)$  and  $b_1(x)$  are the required monic polynomials in  $\mathbb{Z}[x]$ .

**Case 2** If  $w > 1$ , there exists a prime  $p$  such that  $p \mid w$ . Since  $\gcd(v, w) = 1$ , we know  $p \nmid v$ . Consider the ring homomorphism  $\varphi : \mathbb{Z}[x] \rightarrow \mathbb{Z}_p[x]$  defined in Lemma 2.2.7 by  $\varphi(h(x)) = \overline{h(x)}$ .

Applying  $\varphi$  to both sides of Equation (1):

Left Side: Since  $p \mid w$ , we have  $\varphi(wf(x)) = \overline{wf(x)} = 0$  in  $\mathbb{Z}_p[x]$ .

Right Side: By the properties of the ring homomorphism from Lemma 2.2.7:

$$\varphi(v) \cdot \overline{(a_r x^r + \cdots + a_0)} \cdot \overline{(b_s x^s + \cdots + b_0)}$$

Since  $p \nmid v$ ,  $\varphi(v) \neq 0$  in  $\mathbb{Z}_p$ . Furthermore, because  $\gcd(a_r, \dots, a_0) = 1$  and  $\gcd(b_s, \dots, b_0) = 1$ , their images under  $\varphi$  are non-zero in  $\mathbb{Z}_p[x]$ . Since  $\mathbb{Z}_p[x]$  is an integral domain, the product of these non-zero elements must be non-zero:

$$\varphi(v) \cdot \overline{a(x)} \cdot \overline{b(x)} \neq 0$$

This contradicts the fact that the left side is 0. Therefore,  $w$  must be 1, which completes the proof.  $\square$

**Definition 2.2.5** (Minimal Polynomial). Let  $\alpha \in \mathbb{C}$ . The **minimal polynomial** of  $\alpha$  over  $\mathbb{Q}$  is the unique monic polynomial  $p(x) \in \mathbb{Q}[x]$  of least degree such that  $p(\alpha) = 0$ .

**Theorem 2.2.9.** Let  $\alpha \in \mathbb{C}$ , and let  $p(x)$  be its minimal polynomial over  $\mathbb{Q}$ . Then  $p(x)$  is irreducible in  $\mathbb{Q}[x]$ .

*Proof.* Let  $p(x)$  be the minimal polynomial. Suppose that  $p(x)$  is reducible in  $\mathbb{Q}[x]$ . Then there exist nonconstant polynomials  $r(x), s(x) \in \mathbb{Q}[x]$  such that  $p(x) = r(x)s(x)$ . Since  $p(\alpha) = 0$ , we have  $r(\alpha)s(\alpha) = 0$ , thus  $r(\alpha) = 0$  or  $s(\alpha) = 0$ . Without loss of generality, assume that  $r(\alpha) = 0$ . Then  $r(x)$  is a nonzero polynomial having  $\alpha$  as a root. But  $\deg(r) < \deg(p)$ , because  $p(x) = r(x)s(x)$  and  $s(x)$  is nonconstant. This contradicts the definition of  $p(x)$  as the monic polynomial of least degree in  $\mathbb{Q}[x]$  having  $\alpha$  as a root. Therefore,  $p(x)$  is irreducible in  $\mathbb{Q}[x]$ .  $\square$

**Lemma 2.2.10.** Let  $\alpha \in \mathbb{C}$  and  $p(x)$  be its minimal polynomial over  $\mathbb{Q}$ . Then the set  $I = \{f(x) \in \mathbb{Q}[x] : f(\alpha) = 0\}$  is an ideal of  $\mathbb{Q}[x]$  generated by  $p(x)$ ; that is,  $I = \langle p(x) \rangle$ .

*Proof.* First, we show  $I$  is an ideal of  $\mathbb{Q}[x]$ . If  $f, g \in I$ , then  $f(\alpha) - g(\alpha) = 0 - 0 = 0$ , so  $f - g \in I$ . If  $f \in I$  and  $h$  is any polynomial, then  $f(\alpha)h(\alpha) = 0 \cdot h(\alpha) = 0$ , so  $f \cdot h \in I$ . So,  $I$  is an ideal.

We know from Theorem 1.2.11 that in  $\mathbb{Q}[x]$ , the element with the smallest degree in an ideal is the one that generates it. Since  $p(x)$  is the polynomial with the smallest degree that gives  $p(\alpha) = 0$ , it must be the generator. Therefore,  $I = \langle p(x) \rangle$ .  $\square$

**Definition 2.2.6.** Let  $K$  be a subfield of  $\mathbb{C}$ . The set  $\mathcal{O}_K$ , denoted as the set of integral elements of  $K = \mathbb{Q}(\sqrt{d})$  over  $\mathbb{Z}$ , is defined as :

$$\mathcal{O}_K = \{\alpha \in K : \alpha \text{ is integral over } \mathbb{Z}\}.$$

**Definition 2.2.7.** Let  $K = \mathbb{Q}(\sqrt{d})$ . The conjugation map  $c : K \rightarrow K$  is defined by  $c(a + b\sqrt{d}) = a - b\sqrt{d}$ . The element  $c(\alpha)$ , also denoted by  $\alpha^c$ , is called a conjugate of  $\alpha$ .

**Remark 2.2.3.** To understand where this definition comes from, we look at the field automorphisms  $c : K \rightarrow K$ :

1. Any such map fixes the rational numbers, meaning  $\forall x \in \mathbb{Q}, c(x) = x$ .
2. Since  $(\sqrt{d})^2 = d$ , applying the map gives  $c(\sqrt{d})^2 = c(d) = d$ . This means a root must be mapped to a root, so  $c(\sqrt{d}) = \pm\sqrt{d}$ .
3. Therefore, if  $z = a + b\sqrt{d}$ , there are exactly two images (conjugates) of  $z$  under these maps:

$$c(z) = z \quad \text{or} \quad c(z) = a - b\sqrt{d}$$

**Definition 2.2.8.** Let  $\alpha \in \mathbb{Q}(\sqrt{d})$ . The **norm** of  $\alpha$  is a map from  $\mathbb{Q}(\sqrt{d})$  to  $\mathbb{Q}$  defined by:

$$N(\alpha) = \alpha\alpha^c$$

**Theorem 2.2.11.** Let  $\alpha = a + b\sqrt{d} \in K$  with  $a, b \in \mathbb{Q}$ . If  $b \neq 0$ , then the minimal polynomial of  $\alpha$  over  $\mathbb{Q}$  is:

$$p(x) = x^2 - (\alpha + \alpha^c)x + \alpha\alpha^c.$$

*Proof.* Since  $\alpha \notin \mathbb{Q}$  ( $b \neq 0$ ), we are sure to have the minimal monic polynomial with degree at least 2 for  $\alpha$ . We construct a monic polynomial with roots  $\alpha$  and  $\alpha^c$

$$p(x) = (x - \alpha)(x - \alpha^c)$$

Expanding this product gives:

$$\begin{aligned} p(x) &= x^2 - \alpha^c x - \alpha x + \alpha\alpha^c \\ &= x^2 - (\alpha + \alpha^c)x + \alpha\alpha^c \end{aligned}$$

To ensure  $p(x) \in \mathbb{Q}[x]$ , we check the coefficients  $\alpha + \alpha^c = (a + b\sqrt{d}) + (a - b\sqrt{d}) = 2a \in \mathbb{Q}$  and  $\alpha\alpha^c = (a + b\sqrt{d})(a - b\sqrt{d}) = a^2 - db^2 \in \mathbb{Q}$ . Since the coefficients are rational and  $p(\alpha) = 0$ ,  $p(x)$  is the minimal polynomial of  $\alpha$  over  $\mathbb{Q}$ .  $\square$

**Theorem 2.2.12.** If  $\alpha = a + b\sqrt{d} \in \mathcal{O}_K$ , then the minimal polynomial  $p(x)$  of  $\alpha$  over  $\mathbb{Q}$  satisfies  $\alpha + \alpha^c \in \mathbb{Z}$  and  $\alpha\alpha^c \in \mathbb{Z}$ .

*Proof.* Since  $\alpha \in \mathcal{O}_K$ , there exists a monic polynomial  $f(x) \in \mathbb{Z}[x]$  such that  $f(\alpha) = 0$ . In  $\mathbb{Q}[x]$ , we can write  $f(x) = q(x)p(x)$  by Lemma 2.2.10, where  $p(x)$  is the minimal polynomial of  $\alpha$ .

By Lemma 2.2.8, since  $f(x)$  is monic in  $\mathbb{Z}[x]$ , it can be factored into monic polynomials in  $\mathbb{Z}[x]$ . Thus:

$$f(x) = q_1(x)p_1(x) \quad \text{where } q_1, p_1 \in \mathbb{Z}[x]$$

and  $\deg(p_1) = \deg(p)$ .

Since  $f(\alpha) = q_1(\alpha)p_1(\alpha) = 0$  and  $\mathbb{C}$  is an integral domain, then either  $p_1(\alpha) = 0$  or  $q_1(\alpha) = 0$ :

**Case 1:** If  $p_1(\alpha) = 0$ , then since  $p_1(x) \in \mathbb{Z}[x]$  and has the same degree as the minimal monic polynomial  $p(x)$ , it must be that  $p(x) = p_1(x)$ . Therefore,  $p(x) \in \mathbb{Z}[x]$ .

**Case 2:** If  $q_1(\alpha) = 0$ , then  $q_1(x) \in \langle p(x) \rangle$  by Lemma 2.2.10 (because  $p(x)$  is the smallest degree polynomial that gives zero). By applying Lemma 2.2.8 again, we factor  $q_1(x)$  and repeat the process.

This process must stop after at most  $n$  steps (where  $n = \deg f$ ). Eventually, we find that the minimal polynomial  $p(x)$  must have integer coefficients, so  $p(x) \in \mathbb{Z}[x]$ . Now,

1. If  $b \neq 0$ , the minimal polynomial in  $\mathbb{Q}[x]$  is:

$$p(x) = (x - \alpha)(x - \alpha^c) = x^2 - (\alpha + \alpha^c)x + \alpha\alpha^c$$

Since we proved that  $p(x) \in \mathbb{Z}[x]$ , the coefficients must be integers:

$$(\alpha + \alpha^c) \in \mathbb{Z} \quad \text{and} \quad (\alpha\alpha^c) \in \mathbb{Z}$$

2. If  $b = 0$ , then  $\alpha \in \mathbb{Q}$ . Its minimal polynomial in  $\mathbb{Q}[x]$  is  $p(x) = x - \alpha$ . Since  $p(x) \in \mathbb{Z}[x]$ , then  $\alpha \in \mathbb{Z}$ , which means  $\alpha + \alpha^c = 2a \in \mathbb{Z}$  and  $\alpha\alpha^c = a^2 \in \mathbb{Z}$ .

□

**Example 2.2.9.** Let  $\omega = \frac{1+\sqrt{d}}{2}$  where  $d$  is a square-free integer. Then  $\omega \in \mathcal{O}_K$  if and only if  $d \equiv 1 \pmod{4}$ . Since if  $\omega \in \mathcal{O}_K$ , we find its minimal polynomial over  $\mathbb{Q}$ . Setting  $a = 1/2$ ,  $b = 1/2$  we get

$$p(x) = x^2 - \left(2 \cdot \frac{1}{2}\right)x + \left(\left(\frac{1}{2}\right)^2 - d\left(\frac{1}{2}\right)^2\right) = x^2 - x + \frac{1-d}{4}.$$

For  $\omega \in \mathcal{O}_K$ , the coefficients of its minimal polynomial must be integers. Since the leading coefficient is 1 and the coefficient of  $x$  is  $-1$ , we only need the constant term to be an integer:

$$\frac{1-d}{4} \in \mathbb{Z} \text{ if and only if } 1-d = 4k \text{ if and only if } d \equiv 1 \pmod{4}$$

**Theorem 2.2.13.** Let  $d$  be a square free integer. If  $d \equiv 2$  or  $3 \pmod{4}$ , then  $\mathcal{O}_K = \mathbb{Z}[\sqrt{d}]$ , where

$$\mathbb{Z}[\sqrt{d}] = \{a + b\sqrt{d} : a, b \in \mathbb{Z}\}.$$

*Proof.* To show that  $\mathbb{Z}[\sqrt{d}] \subseteq \mathcal{O}_K$ : Let  $\alpha = a + b\sqrt{d} \in \mathbb{Z}[\sqrt{d}]$ . Consider the polynomial:

$$p(x) = x^2 - 2ax + (a^2 - b^2d)$$

Since  $a, b \in \mathbb{Z}$ , then  $2a \in \mathbb{Z}$  and  $(a^2 - b^2d) \in \mathbb{Z}$ , so  $p(x) \in \mathbb{Z}[x]$ . Since  $p(\alpha) = 0$  and  $p(x)$  is a monic polynomial,  $\alpha$  is an integral element over  $\mathbb{Z}$ . Thus,  $\alpha = a + b\sqrt{d} \in \mathcal{O}_K$ .

To show that  $\mathcal{O}_K \subseteq \mathbb{Z}[\sqrt{d}]$ : Let  $\alpha = a + b\sqrt{d} \in \mathcal{O}_K$ . By Theorem 2.2.12,  $\alpha + \alpha^c = 2a \in \mathbb{Z}$  and  $\alpha\alpha^c = a^2 - b^2d \in \mathbb{Z}$ . Write  $a = \frac{x}{2}$  for  $x \in \mathbb{Z}$  and  $b = \frac{n}{m}$  where  $n, m \in \mathbb{Z}, m \neq 0$  and  $\gcd(n, m) = 1$ . Since  $4b^2d = \frac{4d}{m^2}n^2 \in \mathbb{Z}$  and  $\gcd(n^2, m^2) = 1$ , we have  $m^2 | 4d$ . Now, consider the following cases:

**Case 1 :**  $\gcd(m, 2) = 1$ . Then  $m^2 | d$ . Since  $d$  is square-free,  $m = \pm 1$ . Therefore,  $b \in \mathbb{Z}$ . Since  $a^2 - b^2d \in \mathbb{Z}$  and  $b^2d \in \mathbb{Z}$ , we get  $a^2 \in \mathbb{Z}$ . But  $a = \frac{x}{2}$ ,  $x$  is even, so  $a \in \mathbb{Z}$ . Thus,  $\alpha \in \mathbb{Z}[\sqrt{d}]$ .

**Case 2 :**  $\gcd(m, 2) = 2$ . Then  $m = \pm 2$ .

**Subcase 1 :**  $d \equiv 2 \pmod{4}$ . Using  $a^2 - b^2d \in \mathbb{Z}$  we get  $\frac{x^2 - n^2d}{4} \in \mathbb{Z}$ . Since  $2 | d$  and  $4 | x^2 - n^2d$ , then  $x^2$  must be even (i.e.  $2 | x^2$ ) so  $2 | x$ , implying  $a \in \mathbb{Z}$ . This leads to  $b^2d \in \mathbb{Z}$  to get  $\frac{n^2d}{4} \in \mathbb{Z}$ . Since  $d$  is square-free and  $2 | d$ , we have  $2 | n$ , contradicting  $\gcd(n, m) = 1$ .

**Subcase 2 :**  $d \equiv 3 \pmod{4}$  :  $a^2 - b^2d = \frac{x^2 - n^2d}{4} \in \mathbb{Z}$ . For  $d \equiv 3 \equiv -1 \pmod{4}$ , we get  $x^2 + n^2 \equiv 0 \pmod{4}$ . This only happens if  $x$  and  $n$  are both even, contradicting  $\gcd(n, m) = 1$ .

Hence,  $m$  must be  $\pm 1$ , and  $\mathcal{O}_K = \mathbb{Z}[\sqrt{d}]$ . □

**Theorem 2.2.14.** Let  $K = \mathbb{Q}(\sqrt{d})$  and  $d \equiv 2$  or  $3 \pmod{4}$ . Then  $\mathcal{O}_K$  is a subring of  $K$ .

*Proof.* Let  $x, y \in \mathcal{O}_K$ . By Theorem 2.2.13,  $\mathcal{O}_K = \mathbb{Z}[\sqrt{d}]$ , so  $x = a_1 + b_1\sqrt{d}$  and  $y = a_2 + b_2\sqrt{d}$  with  $a_1, b_1, a_2, b_2 \in \mathbb{Z}$ .

First,  $x - y = (a_1 - a_2) + (b_1 - b_2)\sqrt{d}$ . Since  $a_1 - a_2 \in \mathbb{Z}$  and  $b_1 - b_2 \in \mathbb{Z}$ , we have  $x - y \in \mathcal{O}_K$ .

Also,  $xy = (a_1 + b_1\sqrt{d})(a_2 + b_2\sqrt{d}) = (a_1a_2 + b_1b_2d) + (a_1b_2 + b_1a_2)\sqrt{d}$ . Since  $d \in \mathbb{Z}$ ,  $xy \in \mathcal{O}_K$ . □

**Example 2.2.10.** The following are some examples of quadratic fields and their rings of integers when  $d \equiv 2$  or  $3 \pmod{4}$ .

1. **For  $d = 39$ :**

$$K = \mathbb{Q}(\sqrt{39}) = \{a + b\sqrt{39} : a, b \in \mathbb{Q}\},$$

$$\text{and } \mathcal{O}_K = \mathbb{Z}[\sqrt{39}] = \{a + b\sqrt{39} : a, b \in \mathbb{Z}\}.$$

2. **For  $d = 19$ :**

$$K = \mathbb{Q}(\sqrt{19}) = \{a + b\sqrt{19} : a, b \in \mathbb{Q}\},$$

$$\text{and } \mathcal{O}_K = \mathbb{Z}[\sqrt{19}] = \{a + b\sqrt{19} : a, b \in \mathbb{Z}\}.$$

3. **For**  $d = -1$ :

$$K = \mathbb{Q}(\sqrt{-1}) = \{a + b\sqrt{-1} : a, b \in \mathbb{Q}\},$$

$$\text{and } \mathcal{O}_K = \mathbb{Z}[\sqrt{-1}] = \{a + b\sqrt{-1} : a, b \in \mathbb{Z}\}.$$

4. **For**  $d = -14$ :

$$K = \mathbb{Q}(\sqrt{-14}) = \{a + b\sqrt{-14} : a, b \in \mathbb{Q}\},$$

$$\text{and } \mathcal{O}_K = \mathbb{Z}[\sqrt{-14}] = \{a + b\sqrt{-14} : a, b \in \mathbb{Z}\}.$$

**Theorem 2.2.15.** Let  $d \neq 1$  be a square free integer. If  $d \equiv 1 \pmod{4}$ , then  $\mathcal{O}_K = \mathbb{Z}\left[\frac{1+\sqrt{d}}{2}\right]$ , where

$$\mathbb{Z}\left[\frac{1+\sqrt{d}}{2}\right] = \left\{a + b\left(\frac{1+\sqrt{d}}{2}\right) : a, b \in \mathbb{Z}\right\}$$

*Proof.* First, we show that  $\mathbb{Z}\left[\frac{1+\sqrt{d}}{2}\right] \subseteq \mathcal{O}_K$ : Let  $\alpha = a + b\left(\frac{1+\sqrt{d}}{2}\right) \in \mathbb{Z}\left[\frac{1+\sqrt{d}}{2}\right]$ . Then  $\alpha = (a + \frac{b}{2}) + \frac{b}{2}\sqrt{d}$ , where  $a, b \in \mathbb{Z}$ . Hence,  $\alpha + \alpha^c = 2a + b \in \mathbb{Z}$ . Also,  $N(\alpha) = (a + \frac{b}{2})^2 - d(\frac{b}{2})^2 = a^2 + ab + \frac{b^2(1-d)}{4}$ . Since  $d \equiv 1 \pmod{4}$ , we have  $4|(1-d)$  and hence  $N(\alpha) \in \mathbb{Z}$ . Since,  $p(x) = x^2 - (\alpha + \alpha^c)x + \alpha\alpha^c$  is monic in  $\mathbb{Z}[x]$ , and  $p(\alpha) = 0$ , we get  $\alpha \in \mathcal{O}_K$ .

To show that  $\mathcal{O}_K \subseteq \mathbb{Z}\left[\frac{1+\sqrt{d}}{2}\right]$ : Assume that  $a + b\sqrt{d} \in \mathcal{O}_K$  with  $a = \frac{x}{2}$  and  $b = \frac{n}{m}$ ,  $\gcd(n, m) = 1$ . Since  $4b^2d = \frac{4d}{m^2}n^2 \in \mathbb{Z}$ , we get  $m^2|4d$ . But  $\gcd(n^2, m^2) = 1$ , so we have the following cases :

**Case 1 :** If  $m = \pm 1$ : Then  $b \in \mathbb{Z}$  and  $a \in \mathbb{Z}$ . Therefore,  $a + b\sqrt{d} = (a - b) + 2b\left(\frac{1+\sqrt{d}}{2}\right) \in \mathbb{Z}\left[\frac{1+\sqrt{d}}{2}\right]$ .

**Case 2 :** If  $m = \pm 2$ , then  $n$  must be odd since  $\gcd(n, m) = 1$ . Hence  $b = \frac{n}{2}$  and

$$a^2 - b^2d = \frac{x^2 - n^2d}{4} \in \mathbb{Z}, \text{ Thus}$$

$$x^2 - n^2d \equiv 0 \pmod{4} \quad (2)$$

Since  $d \equiv 1 \pmod{4}$  and the square of any integer is congruent to either 0 or 1 modulo 4, the condition  $\gcd(n, 2) = 1$  (i.e.,  $n$  is odd) implies that  $n^2 \equiv 1 \pmod{4}$ . So,  $n^2d \equiv (1)(1) \equiv 1 \pmod{4}$ . Substituting this in (2), we get  $x^2 - 1 \equiv 0 \pmod{4}$ , which means  $x^2 \equiv 1 \pmod{4}$ . Therefore  $x$  must be odd. We can write:

$$a + b\sqrt{d} = \frac{x}{2} + \frac{n}{2}\sqrt{d} = \frac{x - n}{2} + n\left(\frac{1 + \sqrt{d}}{2}\right)$$

Since  $x$  and  $n$  are both odd, we get  $\frac{x-n}{2} \in \mathbb{Z}$ . Thus,  $a + b\sqrt{d} \in \mathbb{Z}\left[\frac{1+\sqrt{d}}{2}\right]$ . □

**Theorem 2.2.16.** Let  $K = \mathbb{Q}(\sqrt{d})$ , where  $d \equiv 1 \pmod{4}$ . Then  $\mathcal{O}_K$  is a subring of  $K$ .

*Proof.* Let  $x, y \in \mathcal{O}_K$ . By Theorem 2.2.15, we have  $x = a_1 + b_1(\frac{1+\sqrt{d}}{2})$  and  $y = a_2 + b_2(\frac{1+\sqrt{d}}{2})$  where  $a_1, b_1, a_2, b_2 \in \mathbb{Z}$ . Then

$$x - y = (a_1 - a_2) + (b_1 - b_2) \left( \frac{1 + \sqrt{d}}{2} \right) \in \mathcal{O}_K$$

$$\text{Now, } xy = a_1a_2 + (a_1b_2 + b_1a_2) \left( \frac{1 + \sqrt{d}}{2} \right) + b_1b_2 \left( \frac{1 + \sqrt{d}}{2} \right)^2$$

Since  $\left( \frac{1+\sqrt{d}}{2} \right)^2 = \frac{1+d+2\sqrt{d}}{4} = \frac{d-1}{4} + \frac{1+\sqrt{d}}{2}$ , and  $d \equiv 1 \pmod{4}$ , we have  $\frac{d-1}{4} \in \mathbb{Z}$ .

Therefore,  $xy = (a_1a_2 + b_1b_2\frac{d-1}{4}) + (a_1b_2 + b_1a_2 + b_1b_2) \left( \frac{1+\sqrt{d}}{2} \right) \in \mathcal{O}_K$ .  $\square$

**Remark 2.2.4.** The subring  $\mathcal{O}_K$  of  $K$  is called the ring of integers of  $K$ .

**Remark 2.2.5.** Let  $d$  be a square-free integer such that  $d \equiv 1 \pmod{4}$ . The quadratic ring  $\mathbb{Z}[\sqrt{d}]$  is a subring of  $\mathbb{Z} \left[ \frac{1+\sqrt{d}}{2} \right]$ .

To see this, note that:

$$\sqrt{d} = -1 + 2 \left( \frac{1 + \sqrt{d}}{2} \right).$$

Hence, every linear combination  $x + y\sqrt{d}$  for  $x, y \in \mathbb{Z}$  must also be in  $\mathbb{Z} \left[ \frac{1+\sqrt{d}}{2} \right]$ , since  $\mathcal{O}_K$  is a ring.

**Example 2.2.11.** The following are some examples of quadratic fields and their rings of integers when  $d \equiv 1 \pmod{4}$  :

1. **For  $d = 5$ :**

$$K = \mathbb{Q}(\sqrt{5}) = \{a + b\sqrt{5} : a, b \in \mathbb{Q}\},$$

$$\text{and } \mathcal{O}_K = \mathbb{Z} \left[ \frac{1+\sqrt{5}}{2} \right] = \{a + b(\frac{1+\sqrt{5}}{2}) : a, b \in \mathbb{Z}\}.$$

2. **For  $d = 13$ :**

$$K = \mathbb{Q}(\sqrt{13}) = \{a + b\sqrt{13} : a, b \in \mathbb{Q}\}.$$

$$\text{and } \mathcal{O}_K = \mathbb{Z} \left[ \frac{1+\sqrt{13}}{2} \right] = \{a + b(\frac{1+\sqrt{13}}{2}) : a, b \in \mathbb{Z}\}.$$

3. **For  $d = -7$ :**

$$K = \mathbb{Q}(\sqrt{-7}) = \{a + b\sqrt{-7} : a, b \in \mathbb{Q}\}.$$

$$\text{and } \mathcal{O}_K = \mathbb{Z} \left[ \frac{1+\sqrt{-7}}{2} \right] = \{a + b(\frac{1+\sqrt{-7}}{2}) : a, b \in \mathbb{Z}\}.$$

**Theorem 2.2.17.** Let  $\alpha, \beta \in K$ . Then:

$$1. (\alpha + \beta)^c = \alpha^c + \beta^c.$$

2.  $(\alpha\beta)^c = \alpha^c\beta^c$ .
3.  $N(\alpha\beta) = N(\alpha)N(\beta)$ .
4.  $N(\alpha^c) = N(\alpha)$ .
5. Every  $q \in \mathbb{Q}$ ,  $N(q) = q^2$ .
6.  $N(\alpha) = 0$  if and only if  $\alpha = 0$ .  
Moreover, if  $\alpha \in \mathcal{O}_K$ , then:
7.  $N(\alpha) \in \mathbb{Z}$ .
8.  $\alpha$  is a unit in  $\mathcal{O}_K$  if and only if  $N(\alpha) = \pm 1$ .

*Proof.* Let  $\alpha = a_1 + b_1\sqrt{d}$  and  $\beta = a_2 + b_2\sqrt{d}$  where  $a_1, b_1, a_2, b_2 \in \mathbb{Q}$ .

1. Since  $\alpha + \beta = (a_1 + a_2) + (b_1 + b_2)\sqrt{d}$ , we get  $(\alpha + \beta)^c = (a_1 + a_2) - (b_1 + b_2)\sqrt{d}$ .  
So,  

$$(\alpha + \beta)^c = (a_1 - b_1\sqrt{d}) + (a_2 - b_2\sqrt{d}) = \alpha^c + \beta^c.$$
2. Since  $\alpha\beta = (a_1a_2 + b_1b_2d) + (a_1b_2 + b_1a_2)\sqrt{d}$ , we have  $(\alpha\beta)^c = (a_1a_2 + b_1b_2d) - (a_1b_2 + b_1a_2)\sqrt{d}$ . Also,

$$\begin{aligned}\alpha^c\beta^c &= (a_1 - b_1\sqrt{d})(a_2 - b_2\sqrt{d}) \\ &= a_1a_2 - a_1b_2\sqrt{d} - b_1a_2\sqrt{d} + b_1b_2d \\ &= (a_1a_2 + b_1b_2d) - (a_1b_2 + b_1a_2)\sqrt{d}\end{aligned}$$

So,  $(\alpha\beta)^c = \alpha^c \cdot \beta^c$ .

3.  $N(\alpha\beta) = N(\alpha)N(\beta)$ :  
Using part(2),  $(\alpha\beta)^c = \alpha^c\beta^c$ . Therefore,  $N(\alpha\beta) = (\alpha\beta)(\alpha\beta)^c = (\alpha\beta)(\alpha^c\beta^c)$ .  
Since  $K$  is commutative, we can rearrange the terms as follows:

$$N(\alpha\beta) = (\alpha\alpha^c)(\beta\beta^c) = N(\alpha)N(\beta)$$

.

4. Since  $(\alpha^c)^c = \alpha$ , we have :

$$N(\alpha^c) = \alpha^c \cdot \alpha = \alpha\alpha^c = N(\alpha)$$

5. For any  $q \in \mathbb{Q}$ ,  $q = q + 0\sqrt{d} \in K$ .

So,

$$N(q) = (q + 0\sqrt{d})(q - 0\sqrt{d}) = q \cdot q = q^2$$

6. If  $\alpha = 0$ , then  $N(0) = 0 \cdot 0^c = 0$ . Conversely, if  $N(\alpha) = 0$ , then  $\alpha\alpha^c = 0$ . Since  $K$  is a field, this implies  $\alpha = 0$  or  $\alpha^c = 0$ . If  $\alpha^c = 0$ , then its conjugate  $(\alpha^c)^c = 0^c$ , so  $\alpha = 0$ . Thus,  $\alpha = 0$  in both cases.

7. Let  $\alpha \in \mathcal{O}_K$ . Then we have the following two cases

**Case 1:**  $d \equiv 2, 3 \pmod{4}$ .

Then  $\alpha = a + b\sqrt{d}$  where  $a, b \in \mathbb{Z}$ , so  $N(\alpha) = a^2 - b^2d \in \mathbb{Z}$

**Case 2:**  $d \equiv 1 \pmod{4}$ .

Then  $\alpha = a + b\left(\frac{1+\sqrt{d}}{2}\right) = \left(a + \frac{b}{2}\right) + \frac{b}{2}\sqrt{d}$ , where  $a, b \in \mathbb{Z}$ . So,

$$N(\alpha) = \left(a + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2 d = a^2 + ab + \frac{b^2}{4} - \frac{b^2d}{4}$$

$$N(\alpha) = a^2 + ab + \frac{b^2(1-d)}{4}.$$

Since  $d \equiv 1 \pmod{4}$ , we get  $\frac{1-d}{4} \in \mathbb{Z}$ . Thus,  $N(\alpha) \in \mathbb{Z}$ .

8. Suppose  $\alpha$  is a unit in  $\mathcal{O}_K$ . Then  $\alpha\beta = 1$  for some  $\beta \in \mathcal{O}_K$ . Taking norms and using part(3) to get  $N(\alpha)N(\beta) = 1$ . Since  $N(\alpha), N(\beta) \in \mathbb{Z}$ , we have  $N(\alpha) = \pm 1$ . Conversely, suppose that  $N(\alpha) = \pm 1$ . If  $N(\alpha) = 1$ , then so  $\alpha^{-1} = \alpha^c$ . If  $N(\alpha) = -1$ , then  $\alpha(-\alpha^c) = 1$ , so  $\alpha^{-1} = -\alpha^c$ . Since  $\alpha \in \mathcal{O}_K$ , we have  $\pm\alpha^c \in \mathcal{O}_K$ . Therefore  $\alpha$  has an inverse in  $\mathcal{O}_K$ , so  $\alpha$  is a unit.

□

**Lemma 2.2.18.** An element  $\alpha \in \mathcal{O}_K$  is irreducible if and only if its conjugate  $\alpha^c$  is irreducible.

*Proof.* Let  $\alpha^c = \beta\gamma$  where  $\beta, \gamma \in \mathcal{O}_K$ . If  $\alpha$  is irreducible in  $\mathcal{O}_K$ , then  $\alpha = (\beta\gamma)^c = \beta^c\gamma^c$ . Since if  $\beta^c$  or  $\gamma^c$  is a unit, then  $\beta$  or  $\gamma$  is a unit respectively, we have  $\alpha^c$  is irreducible.

Now for the converse,  $\alpha = \beta\gamma$  where  $\beta, \gamma \in \mathcal{O}_K$ . If  $\alpha^c$  is irreducible in  $\mathcal{O}_K$ , then  $\alpha^c = (\beta\gamma)^c = \beta^c\gamma^c$ . Since if  $\beta^c$  or  $\gamma^c$  is a unit, then  $\beta$  or  $\gamma$  is a unit respectively, we have  $\alpha$  is irreducible.

□

**Remark 2.2.6.**  $\mathcal{O}_K$  is not a field, by Theorem 2.2.17 part (8) we have,  $2 \in \mathcal{O}_K$  but  $N(2) = 4 \neq \pm 1$ .

**Lemma 2.2.19.** Let  $K = \mathbb{Q}(\sqrt{d})$ , where  $d \equiv 1 \pmod{4}$ . If  $\alpha \in \mathcal{O}_K$ , then  $\alpha = \frac{a+b\sqrt{d}}{2} \in \mathcal{O}_K$  with  $a, b \in \mathbb{Z}$ , then  $a \equiv b \pmod{2}$ . In particular,  $a$  and  $b$  have the same parity.

*Proof.* Since  $\alpha \in \mathcal{O}_K = \mathbb{Z} \left[ \frac{1+\sqrt{d}}{2} \right]$ , there exist  $m, n \in \mathbb{Z}$  such that:

$$\alpha = m + n \left( \frac{1 + \sqrt{d}}{2} \right).$$

Thus,  $\alpha = \frac{2m}{2} + \frac{n}{2} + \frac{n\sqrt{d}}{2} = \frac{2m+n}{2} + \frac{n\sqrt{d}}{2}$ . Let  $a = 2m + n$  and  $b = n$ , then  $\frac{a+b\sqrt{d}}{2} = \frac{2m+n+n\sqrt{d}}{2}$ .

Moreover,  $a - b = (2m + n) - n = 2m$ . Since  $2m$  is an even integer, it follows that  $a - b \equiv 0 \pmod{2}$ , and hence  $a \equiv b \pmod{2}$ . Therefore,  $a$  and  $b$  are either both even or both odd.  $\square$

We now compare the behavior of units in imaginary and real quadratic fields. In imaginary quadratic fields, the set of units is finite, while in real quadratic fields there are infinitely many units of norm 1. This difference is related to the norm equation.

**Remark 2.2.7.** Let  $K = \mathbb{Q}(\sqrt{d})$ , where  $d < 0$  is a square-free integer, then the norm of every nonzero element in  $\mathcal{O}_K$  is positive integer. Therefore,  $\alpha$  is unit in  $\mathcal{O}_K$  if and only if  $N(\alpha) = 1$ .

**Theorem 2.2.20** (Units of imaginary quadratic fields). Let  $K = \mathbb{Q}(\sqrt{d})$  be an imaginary quadratic field, where  $d < 0$  is a square-free integer. Then the group of units  $U(\mathcal{O}_K)$  is given by:

$$U(\mathcal{O}_K) = \begin{cases} \{\pm 1, \pm i\}, & \text{if } d = -1, \\ \{\pm 1, \pm \omega, \pm \omega^2\}, & \text{if } d = -3, \\ \{\pm 1\}, & \text{otherwise.} \end{cases}$$

where  $\omega = \frac{-1+\sqrt{-3}}{2}$ .

*Proof.* Let  $u \in \mathcal{O}_K$ . Then we have 2 cases:

**Case 1:**  $d \equiv 2, 3 \pmod{4}$ . Here,  $u = x + y\sqrt{d}$  for  $x, y \in \mathbb{Z}$ , so  $N(u) = x^2 + |d|y^2 = 1$ .

If  $d = -1$ , then  $x^2 + y^2 = 1$  has solutions  $x = \pm 1, y = 0$  and  $x = 0, y = \pm 1$ ,

For both possible solutions, we will have  $U(\mathcal{O}_K) = \{\pm 1, \pm i\}$ .

If  $d < -1$ , then  $|d| \geq 2$ , so  $x = \pm 1$  and  $y = 0$ . So  $U(\mathcal{O}_K) = \{\pm 1\}$ .

**Case 2:**  $d \equiv 1 \pmod{4}$ . Here, by Lemma 2.2.19  $u = \frac{x+y\sqrt{d}}{2}$  for  $x, y \in \mathbb{Z}$  with  $x \equiv y \pmod{2}$ .

So,  $N(u) = \frac{x^2 - dy^2}{4} = \frac{x^2 + |d|y^2}{4} = 1$ . Multiply both sides by 4 to get  $x^2 + |d|y^2 = 4$ .  
Now

For  $d = -3$ , the possible integer solutions for  $x^2 + 3y^2 = 4$  with  $x \equiv y \pmod{2}$  are  $(x, y) = (\pm 2, 0), (\pm 1, \pm 1)$ . If  $(x, y) = (\pm 2, 0)$ , then  $u \in \{\pm 1\}$  and if  $(x, y) = (\pm 1, \pm 1)$ , then  $u \in \{\frac{\pm 1 \pm \sqrt{-3}}{2}\}$ . Therefore,  $U(\mathcal{O}_K) = \{\pm 1, \pm \omega, \pm \omega^2\}$  where  $\omega = \frac{-1 + \sqrt{-3}}{2}$ .

For  $d < -3$ , we have  $|d| \geq 7$  and we must have  $(x, y) = (\pm 2, 0)$  as a solution for  $x^2 + |d|y^2 = 4$ . Thus,  $U(\mathcal{O}_K) = \{\pm 1\}$ .

□

**Theorem 2.2.21.** Let  $K = \mathbb{Q}(\sqrt{d})$  be a real quadratic field. Then there exist infinitely many units  $u \in \mathcal{O}_K$  with  $N(u) = 1$ .

*Proof.* Since  $d > 0$  is a square-free integer, we have by Theorem 1.1.21 that, Pell's equation  $x^2 - dy^2 = 1$  has infinitely many distinct solutions in positive integers. Each solution produces an element  $u = x + y\sqrt{d}$  with norm  $N(u) = x^2 - dy^2 = 1$ . We verify that these infinitely many elements belong to  $\mathcal{O}_K$  by analyzing its possible cases:

If  $d \equiv 2, 3 \pmod{4}$ , then  $\mathcal{O}_K = \mathbb{Z}[\sqrt{d}]$ . Every integer solution pair  $(x, y)$  directly defines a unique valid unit  $u = x + y\sqrt{d} \in \mathcal{O}_K$  satisfying  $N(u) = 1$ .

If  $d \equiv 1 \pmod{4}$ , then  $\mathcal{O}_K = \mathbb{Z}[\frac{1+\sqrt{d}}{2}]$ . Because  $\mathbb{Z}[\sqrt{d}]$  is contained as a subring within  $\mathcal{O}_K$ , any element of the form  $u = x + y\sqrt{d}$  with  $x, y \in \mathbb{Z}$  remains a valid in  $\mathcal{O}_K$ , preserving the infinitely many solutions satisfying  $N(u) = 1$ .

Thus, in both cases, the Pell's equation guarantees infinitely many distinct units of norm 1 in  $\mathcal{O}_K$ . □

## 2.3 Mersenne Prime Analogue in Real Quadratic Fields

In this section, we study Mersenne primes in real quadratic fields  $K = \mathbb{Q}(\sqrt{d})$  under the assumption that  $\mathcal{O}_K$  is a PID. We follow the definition introduced in the article [7], but we revise and correct some of the stated theorems where necessary. We also provided detailed proofs and additional examples to clarify the results.

**Definition 2.3.1.** Let  $K = \mathbb{Q}(\sqrt{d})$  with  $d > 1$  square-free, and let  $\mathcal{O}_K$  be its ring of integers. According to the existence conditions mentioned later in Theorem 2.3.8 and Theorem 2.3.10 we can choose an irreducible element  $\alpha \in \mathcal{O}_K$  such that  $u = \alpha - 1$  is a unit in  $\mathcal{O}_K$ . For a rational prime  $p > 0$ , define

$$M_{p,\alpha} = \frac{\alpha^p - 1}{\alpha - 1}.$$

We call  $M_{p,\alpha}$  a **Mersenne prime analogue** if  $|N(M_{p,\alpha})|$  is a rational prime.

**Theorem 2.3.1.** Let  $\mathcal{O}_K$  be a principal ideal domain (PID). If  $\alpha \in \mathcal{O}_K$  is an irreducible element, then  $|N(\alpha)| = p$  or  $|N(\alpha)| = p^2$  for some rational prime  $p$ .

*Proof.* Suppose  $\alpha$  is an irreducible element in  $\mathcal{O}_K$ . Since  $\mathcal{O}_K$  is a PID,  $\alpha$  is a non-zero, non-unit prime element in  $\mathcal{O}_K$ . This implies that  $|N(\alpha)| > 1$ .

By the Fundamental Theorem of Arithmetic, we can write:

$$|N(\alpha)| = p_1 p_2 \cdots p_m$$

where  $p_1, p_2, \dots, p_m$  are rational primes.

Since  $N(\alpha) = \alpha \alpha^c$ , we know that  $\alpha$  divides  $N(\alpha)$ , which means  $\alpha \mid |N(\alpha)|$ . Thus,  $\alpha \mid p_1 p_2 \cdots p_m$ . Since  $\alpha$  is a prime element in  $\mathcal{O}_K$ , then  $\alpha \mid p_i$  for some  $i$ .

Therefore, there exists  $\beta \in \mathcal{O}_K$  such that:

$$p_i = \alpha \beta.$$

Taking the norm on both sides, we get:

$$N(p_i) = N(\alpha)N(\beta) \text{ and hence } p_i^2 = N(\alpha)N(\beta).$$

Taking the absolute value gives:

$$p_i^2 = |N(\alpha)||N(\beta)|.$$

Since  $|N(\alpha)| > 1$  and  $|N(\alpha)|$  divides  $p_i^2$ , we must have:

$$|N(\alpha)| = p_i \quad \text{or} \quad |N(\alpha)| = p_i^2$$

where  $p_i$  is a prime integer. □

**Theorem 2.3.2.** Let  $\alpha$  in  $\mathcal{O}_K$ . If  $|N(\alpha)| = q$ , where  $q$  is a rational prime, then  $\alpha$  is irreducible in  $\mathcal{O}_K$ .

*Proof.* Let  $N(\alpha)$  be a rational prime  $q$ . Suppose that  $\alpha = \beta\gamma$  for some  $\beta, \gamma \in \mathcal{O}_K$ . To show that  $\alpha$  is irreducible, we must show that  $\beta$  is a unit or  $\gamma$  is a unit. Taking the norm and the absolute values for both sides to have

$$|N(\alpha)| = |N(\beta)N(\gamma)|$$

Hence

$$q = |N(\beta)||N(\gamma)|.$$

Since  $q$  is a rational prime,

$$|N(\beta)| = 1 \quad \text{or} \quad |N(\gamma)| = 1$$

So  $\beta$  or  $\gamma$  is a unit. Thus  $\alpha$  is irreducible in  $\mathcal{O}_K$ . □

**Lemma 2.3.3.** For any  $\alpha \in \mathcal{O}_K$  and any positive integer  $m$ , the following identity holds:

$$\alpha^m - 1 = (\alpha - 1)(1 + \alpha + \cdots + \alpha^{m-1}). \quad (3)$$

*Proof.* By expanding the product on the right-hand side, we have:

$$\begin{aligned} (\alpha - 1)(1 + \alpha + \cdots + \alpha^{m-1}) &= (\alpha + \alpha^2 + \cdots + \alpha^m) - (1 + \alpha + \cdots + \alpha^{m-1}) \\ &= \alpha + \alpha^2 + \cdots + \alpha^m - 1 - \alpha - \cdots - \alpha^{m-1} \\ &= \alpha^m - 1 \end{aligned}$$

□

**Theorem 2.3.4.** Let  $K = \mathbb{Q}(\sqrt{d})$ , where  $d$  is a square-free integer. If  $\alpha \in \mathcal{O}_K$ , then for any positive integer  $p$ , the element

$$M_{p,\alpha} = \sum_{i=0}^{p-1} \alpha^i$$

is also an element of  $\mathcal{O}_K$ .

*Proof.* Since  $\alpha \in \mathcal{O}_K$  and  $\mathcal{O}_K$  is a subring of  $K$ , it follows by the property of closure under multiplication that  $\alpha^i \in \mathcal{O}_K$  for all  $i \in \{0, 1, \dots, p-1\}$ . Furthermore, because  $\mathcal{O}_K$  is closed under finite addition, the sum of these powers must also be in  $\mathcal{O}_K$ . Thus:

$$M_{p,\alpha} = 1 + \alpha + \alpha^2 + \cdots + \alpha^{p-1} \in \mathcal{O}_K$$

□

**Theorem 2.3.5.** If  $|N(M_{p,\alpha})| = q$ , where  $q$  is a rational prime, then  $M_{p,\alpha}$  is irreducible in  $\mathcal{O}_K$ .

*Proof.* Let  $N(M_{p,\alpha})$  be a rational prime  $q$ . Suppose that  $M_{p,\alpha} = \beta\gamma$  for some  $\beta, \gamma \in \mathcal{O}_K$ . To show that  $M_{p,\alpha}$  is irreducible, we must show that  $\beta$  is a unit or  $\gamma$  is a unit.

Taking the norm and the absolute values for both sides to have

$$|N(M_{p,\alpha})| = |N(\beta)N(\gamma)|$$

Hence

$$q = |N(\beta)||N(\gamma)|.$$

Since  $q$  is a rational prime,

$$|N(\beta)| = 1 \quad \text{or} \quad |N(\gamma)| = 1$$

So  $\beta$  or  $\gamma$  is a unit. Thus  $M_{p,\alpha}$  is irreducible in  $\mathcal{O}_K$ . □

**Note:** For  $1 < d < 100$ , where  $d$  is square free integer,  $\mathcal{O}_K$  is PID exactly for the following, see [4]:

$$d \equiv 1 \pmod{4}: \quad 5, 13, 17, 21, 29, 33, 37, 41, 53, 57, 61, 69, 73, 77, 89, 93, 97.$$

$$d \equiv 2 \pmod{4}: \quad 2, 6, 14, 22, 38, 46, 62, 86, 94.$$

$$d \equiv 3 \pmod{4}: \quad 3, 7, 11, 19, 23, 31, 43, 47, 59, 67, 71, 83.$$

**Example 2.3.2.** Consider the real quadratic field  $K = \mathbb{Q}(\sqrt{2})$ , since  $2 \equiv 2 \pmod{4}$ , the ring of integers  $\mathcal{O}_K = \mathbb{Z}[\sqrt{2}]$ . Now, let  $\alpha = 2 + \sqrt{2} \in \mathcal{O}_K$  and let  $u = \alpha - 1 = 1 + \sqrt{2}$ .

First, we compute the norm of  $u$ :

$$N(u) = (1 + \sqrt{2})(1 - \sqrt{2}) = 1 - 2 = -1$$

Since  $N(u) = -1$ ,  $u$  is a unit in  $\mathcal{O}_K$ . Furthermore,  $1 + \sqrt{2} \neq \pm 1$  so, we have  $u \neq \pm 1$ . Now, we will compute the Mersenne prime analogue  $M_{3,\alpha}$  using two different methods:

(a) Using the expansion  $M_{3,\alpha} = 1 + \alpha + \alpha^2$ :

$$\begin{aligned} M_{3,\alpha} &= 1 + (2 + \sqrt{2}) + (2 + \sqrt{2})^2 \\ &= 3 + \sqrt{2} + (4 + 4\sqrt{2} + 2) \\ &= 9 + 5\sqrt{2} \end{aligned}$$

(b) Using  $M_{3,\alpha} = \frac{\alpha^3-1}{\alpha-1}$ :

$$\begin{aligned}
M_{3,\alpha} &= \frac{(2 + \sqrt{2})^3 - 1}{(2 + \sqrt{2}) - 1} \\
&= \frac{19 + 14\sqrt{2}}{1 + \sqrt{2}} \\
&= \frac{(19 + 14\sqrt{2})(1 - \sqrt{2})}{(1 + \sqrt{2})(1 - \sqrt{2})} \\
&= \frac{19 - 19\sqrt{2} + 14\sqrt{2} - 28}{1 - 2} \\
&= \frac{-9 - 5\sqrt{2}}{-1} \\
&= 9 + 5\sqrt{2}
\end{aligned}$$

Evaluating the norm of  $M_{3,\alpha} = 9 + 5\sqrt{2}$ , we obtain:

$$N(M_{3,\alpha}) = 9^2 - 2(5^2) = 81 - 50 = 31$$

Finally, we check if  $\alpha$  is irreducible, by taking the norm:

$$N(\alpha) = 4 - 2 = 2$$

Since 31 and 2 is a rational prime,  $M_{3,\alpha}$  is a Mersenne prime analogue in  $\mathcal{O}_K$ .

**Theorem 2.3.6.** Let the sequences  $(a_n)$  and  $(b_n)$  be defined by the relation  $\alpha^n = a_n + b_n\sqrt{2}$ , where  $a_n, b_n \in \mathbb{Z}$  and  $\alpha = 2 + \sqrt{2}$ . Then  $(a_n)$  and  $(b_n)$  satisfy the recurrence relations:

$$a_{n+1} = 2a_n + 2b_n, \quad b_{n+1} = a_n + 2b_n$$

with initial values  $a_0 = 1$  and  $b_0 = 0$ .

*Proof.* We proceed by mathematical induction on  $n$ .

**Base Case:** For  $n = 0$ ,  $\alpha^0 = 1 = 1 + 0\sqrt{2}$ . Thus,  $a_0 = 1$  and  $b_0 = 0$ , satisfying the initial conditions.

**Inductive Step:** Assume that for some  $n \geq 0$ ,  $\alpha^n = a_n + b_n\sqrt{2}$  holds. We evaluate  $\alpha^{n+1}$ :

$$\begin{aligned}
\alpha^{n+1} &= \alpha \cdot \alpha^n \\
&= (2 + \sqrt{2})(a_n + b_n\sqrt{2}) \\
&= 2a_n + 2b_n\sqrt{2} + a_n\sqrt{2} + 2b_n \\
&= (2a_n + 2b_n) + (a_n + 2b_n)\sqrt{2}
\end{aligned}$$

By definition,  $\alpha^{n+1} = a_{n+1} + b_{n+1}\sqrt{2}$ . Since:

$$\begin{aligned} a_{n+1} &= 2a_n + 2b_n \\ b_{n+1} &= a_n + 2b_n \end{aligned}$$

we complete the induction proof. So:

$$\alpha^n = a_n + b_n\sqrt{2}, \text{ for all } n.$$

□

**Example 2.3.3.** Applying the recurrence relations in Theorem 2.3.6, when  $\alpha = 2 + \sqrt{2}$  we compute  $\alpha^n$  for  $0 \leq n \leq 4$ :

- $n = 0 : a_0 = 1, b_0 = 0 \implies \alpha^0 = 1$
- $n = 1 : a_1 = 2(1) + 0 = 2, \quad b_1 = 1 + 0 = 1 \implies \alpha^1 = 2 + \sqrt{2}$
- $n = 2 : a_2 = 2(2) + 2(1) = 6, \quad b_2 = 2 + 2(1) = 4 \implies \alpha^2 = 6 + 4\sqrt{2}$
- $n = 3 : a_3 = 2(6) + 2(4) = 20, \quad b_3 = 6 + 2(4) = 14 \implies \alpha^3 = 20 + 14\sqrt{2}$
- $n = 4 : a_4 = 2(20) + 2(14) = 68, \quad b_4 = 20 + 2(14) = 48 \implies \alpha^4 = 68 + 48\sqrt{2}$

We can now construct  $M_{5,\alpha} = \sum_{k=0}^4 \alpha^k$ :

$$\begin{aligned} M_{5,\alpha} &= (1 + 2 + 6 + 20 + 68) + (0 + 1 + 4 + 14 + 48)\sqrt{2} \\ &= 97 + 67\sqrt{2} \end{aligned}$$

The norm is given by:

$$N(M_{5,\alpha}) = 97^2 - 2(67^2) = 9409 - 2(4489) = 9409 - 8978 = 431$$

Since 431 is prime,  $M_{5,\alpha}$  is a Mersenne prime analogue.

**Remark 2.3.1.** Extending these computations, that are in Example 2.3.3, for other prime  $p$ , where  $p \in \{3, 5, 7, 11\}$ , we obtain the following results for  $M_{p,\alpha}$  and its norm. Note that  $N(a + b\sqrt{2}) = a^2 - 2b^2$ .

| Prime ( $p$ ) | $M_{p,\alpha}$            | $N(M_{p,\alpha})$ | Is $N(M_{p,\alpha})$ Prime? |
|---------------|---------------------------|-------------------|-----------------------------|
| $p = 3$       | $9 + 5\sqrt{2}$           | 31                | <b>True</b>                 |
| $p = 5$       | $97 + 67\sqrt{2}$         | 431               | <b>True</b>                 |
| $p = 7$       | $1121 + 791\sqrt{2}$      | 5279              | <b>True</b>                 |
| $p = 11$      | $152193 + 107615\sqrt{2}$ | 732799            | <b>True</b>                 |

**Example 2.3.4.** Consider  $K = \mathbb{Q}(\sqrt{2})$ . Take  $\alpha = 3 + \sqrt{2} \in \mathcal{O}_K$ . Then  $N(\alpha) = 7$  is a prime.

$$\begin{aligned} M_{3,\alpha} &= 1 + \alpha + \alpha^2 \\ &= 1 + (3 + \sqrt{2}) + (3 + \sqrt{2})^2 \\ &= 15 + 7\sqrt{2} \end{aligned}$$

Computing its norm:

$$N(M_{3,\alpha}) = 15^2 - 2(7^2) = 225 - 98 = 127 \text{ (rational prime)}$$

We check the base  $u = \alpha - 1 = 2 + \sqrt{2}$ :

$$N(u) = 2^2 - 2(1^2) = 2 \neq \pm 1$$

Since  $u$  is not a unit in  $\mathcal{O}_K = \mathbb{Z}[\sqrt{2}]$ ,  $M_{3,\alpha}$  is not a Mersenne prime analogue.

**Theorem 2.3.7.** Let  $\alpha$  be irreducible element in  $\mathcal{O}_K$  where  $u = \alpha - 1$  is unit not equal to  $\pm 1$ , and  $p$  be a rational prime. Then

$$N(M_{p,\alpha^c}) = N(M_{p,\alpha})$$

*Proof.* Let:

$$\begin{aligned} M_{p,\alpha} &= 1 + \alpha + \cdots + \alpha^{p-1} \\ M_{p,\alpha^c} &= 1 + \alpha^c + \cdots + (\alpha^c)^{p-1} \end{aligned}$$

Since  $(\alpha^i)^c = (\alpha \cdot \alpha \cdots \alpha)^c = \alpha^c \cdot \alpha^c \cdots \alpha^c = (\alpha^c)^i$ , we get:

$$\begin{aligned} M_{p,\alpha^c} &= 1 + \alpha^c + (\alpha^2)^c + \cdots + (\alpha^{p-1})^c \\ &= (1 + \alpha + \alpha^2 + \cdots + \alpha^{p-1})^c \\ &= (M_{p,\alpha})^c \end{aligned}$$

Therefore,

$$N(M_{p,\alpha^c}) = N((M_{p,\alpha})^c) = N(M_{p,\alpha})$$

□

**Remark 2.3.2.** In the remaining part of this section, we assume that  $K = \mathbb{Q}(\sqrt{d})$  is a real quadratic field, where  $d > 1$ , and the ring of integers  $\mathcal{O}_K$  is a principal ideal domain. Consequently, every irreducible of  $\mathcal{O}_K$  is prime.

**Theorem 2.3.8.** Let  $d \equiv 2, 3 \pmod{4}$  and  $\alpha = u + 1$ , where  $u$  is a unit in  $\mathcal{O}_K$  such that  $N(u) = -1$ . Then  $\alpha$  is irreducible if and only if  $d = 2$  and  $u \in \{a + b\sqrt{2} : a, b \in \{-1, 1\}\}$ .

*Proof.* Let  $\alpha$  be an irreducible element in  $\mathcal{O}_K$  with  $\alpha = u + 1$  and  $u = a + b\sqrt{d}$  a unit in  $\mathcal{O}_K$ .

Then we compute the norm:

$$\begin{aligned} N(\alpha) &= N((a + 1) + b\sqrt{d}) \\ &= (a + 1)^2 - b^2d \\ &= a^2 + 2a + 1 - b^2d \\ &= (a^2 - b^2d) + 2a + 1 \end{aligned}$$

Since  $N(u) = a^2 - b^2d = -1$ , substituting this gives:

$$\begin{aligned} N(\alpha) &= -1 + 2a + 1 \\ &= 2a \end{aligned}$$

Let  $p > 0$ , by Theorem 2.3.1,  $|N(\alpha)| = p$  or  $p^2$ , where  $p$  is a rational prime. Note that,  $2|a| = p$ . Since if  $2|a| = p^2$ , we have  $a = \pm 2$  (assuming  $p = 2$ ,  $p^2 = 4$ ). If  $a = \pm 2$ , then  $d = 5 \not\equiv 2, 3 \pmod{4}$ . This is because  $N(u) = 4 - b^2d = -1 \implies b^2d = 5$ . Now, for  $2|a| = p$ , we have  $a = \pm 1$  with  $b = \pm 1$  and  $d = 2$ . Since  $N(u) = 1 - b^2d = -1 \implies b^2d = 2$ . Therefore, we get  $a, b \in \{-1, 1\}$  and  $d = 2$ . Hence  $u \in \{a + b\sqrt{2} : a, b \in \{-1, 1\}\}$ .

Conversely, Let  $u \in \{a + b\sqrt{2} : a, b \in \{-1, 1\}\}$  and  $\alpha = u + 1$ .

Then  $N(u) = a^2 - b^2(2) = 1 - 2 = -1$ . We only have to show that  $\alpha$  is irreducible. Now, if  $u = -1 \pm \sqrt{2}$ , we have  $\alpha = \pm\sqrt{2}$  with:

$$|N(\alpha)| = |N(\pm\sqrt{2})| = |-2| = 2 \text{ (rational prime)}$$

So  $\alpha$  is irreducible.

And for  $u = 1 \pm \sqrt{2}$ , we have  $\alpha = 2 \pm \sqrt{2}$  with:

$$|N(\alpha)| = |N(2 \pm \sqrt{2})| = |4 - 2| = 2 \text{ (rational prime)}$$

So  $\alpha$  is irreducible, and hence  $\alpha$  is irreducible for all cases. □

According to Theorem 2.2.11, when  $d = 2$ , the possible units are:

$$u \in \{1 + \sqrt{2}, 1 - \sqrt{2}, -1 + \sqrt{2}, -1 - \sqrt{2}\}.$$

Therefore, the corresponding values of  $\alpha = u + 1$  are:

$$\alpha \in \{2 + \sqrt{2}, 2 - \sqrt{2}, \sqrt{2}, -\sqrt{2}\}.$$

Notice that  $2 - \sqrt{2}$  is the conjugate of  $2 + \sqrt{2}$  and  $-\sqrt{2}$  is the conjugate of  $\sqrt{2}$ . Hence by Theorem 2.3.7, conjugate choices of  $\alpha$  produce conjugate Mersenne numbers and have the same norm. Therefore, it is enough to study one representative from each conjugate pair.

In the next two examples, we consider  $\alpha = 2 + \sqrt{2}$  and  $\alpha = \sqrt{2}$ .

**Example 2.3.5.** In  $\mathbb{Q}(\sqrt{2})$ , we generated the following Mersenne primes in  $\mathbb{Q}(\sqrt{2})$  using our own [Mathematica code](#):

| $p$ | $M_{p,\alpha}$            | $N(M_{p,\alpha})$ |
|-----|---------------------------|-------------------|
| 2   | $3 + \sqrt{2}$            | 7                 |
| 3   | $9 + 5\sqrt{2}$           | 31                |
| 5   | $97 + 67\sqrt{2}$         | 431               |
| 7   | $1121 + 791\sqrt{2}$      | 5279              |
| 11  | $152193 + 107615\sqrt{2}$ | 732799            |

Table 2.2: Some Mersenne prime analogies in  $\mathbb{Q}(\sqrt{2})$

The next three Mersenne prime analogues are computationally observed at  $p = 73$ , with

$$N(M_{73,\alpha}) = 851569055172258793218602741480913108991,$$

$p = 89$  with

$$N(M_{89,\alpha}) = 290315886781191681464330388772329064268797313023,$$

and at  $p = 233$  with

$$N(M_{233,\alpha}) = 180604754272820230333680012311664417847378068915378065$$

$$4706531416791195\ 9518498581747712829157156517940837234519177963497324543.$$

The next rational prime that gives a Mersenne prime analogue is  $p = 4057$ , with norm length 2164. We found it by using this [Mathematica code](#).

**Example 2.3.6.** In  $\mathbb{Q}(\sqrt{2})$ , we will take  $\alpha = \sqrt{2}$ .

In this case, the norm of  $M_{p,\alpha}$  is related to the classical Mersenne numbers. Specifically, for every odd rational prime  $p > 1$ , we have:

$$\begin{aligned}
M_{p,\alpha} &= \frac{(\sqrt{2})^p - 1}{\sqrt{2} - 1} \\
&= \frac{-1 + 2^{\frac{p-1}{2}} \cdot \sqrt{2}}{-1 + \sqrt{2}} \\
&= (1 + \sqrt{2}) \left( -1 + 2^{\frac{p-1}{2}} \cdot \sqrt{2} \right)
\end{aligned}$$

Hence,

$$\begin{aligned}
N(M_{p,\alpha}) &= N(1 + \sqrt{2}) \cdot N\left(-1 + 2^{\frac{p-1}{2}} \cdot \sqrt{2}\right) \\
&= -1 \cdot (1 - 2^{p-1} \cdot 2) \\
&= 2^p - 1.
\end{aligned}$$

However, for  $p = 2$ , we have:

$$\begin{aligned}
M_{2,\alpha} &= \frac{1}{\sqrt{2} - 1} \\
&= 1 + \sqrt{2}
\end{aligned}$$

Hence,  $N(M_{2,\alpha}) = N(1 + \sqrt{2}) = -1$ , and thus  $M_{2,\alpha}$  is a unit.

Therefore, all the odd primes that gave classical Mersenne primes in Table 2.1 will give us Mersenne prime analogues.

**Lemma 2.3.9.** Let  $d \equiv 2, 3 \pmod{4}$ , and let  $\mathcal{O}_K = \mathbb{Z}[\sqrt{d}]$ . Then 2 is not prime in  $\mathcal{O}_K$ .

*Proof.* We consider two cases based on the  $d$  modulo 4:

**Case 1:**  $d \equiv 2 \pmod{4}$

Since  $d$  is even, we have  $2 \mid d = (\sqrt{d})(\sqrt{d})$ . Note that  $N(\sqrt{d}) = -d \neq \pm 1$ . If  $2 \mid \sqrt{d}$ , then there exist integers  $a, b$  such that:

$$\sqrt{d} = 2(a + b\sqrt{d}) = 2a + 2b\sqrt{d}.$$

Comparing coefficients, we get  $2a = 0$  and  $2b = 1$ , which is impossible since  $b \in \mathbb{Z}$ . Hence  $2 \nmid \sqrt{d}$ , and therefore 2 is not prime in this case.

**Case 2:**  $d \equiv 3 \pmod{4}$

Since  $d$  is odd,  $1 - d$  is even. We observe that:

$$(1 + \sqrt{d})(1 - \sqrt{d}) = 1 - d.$$

Thus,  $2 \mid (1 + \sqrt{d})(1 - \sqrt{d})$ . We now check if 2 divides either factor. If  $2 \mid (1 + \sqrt{d})$ , then there exist integers  $a, b$  such that:

$$1 + \sqrt{d} = 2(a + b\sqrt{d}) = 2a + 2b\sqrt{d}.$$

This implies  $1 = 2a$  and  $1 = 2b$ , a contradiction as  $a, b \in \mathbb{Z}$ . Similarly,  $2 \nmid (1 - \sqrt{d})$ . Since 2 divides a product but does not divide either factor, 2 is not prime.  $\square$

**Theorem 2.3.10.** Let  $d \equiv 2, 3 \pmod{4}$  and  $\alpha = u + 1$ , where  $u$  is a unit in  $\mathcal{O}_K$  such that  $N(u) = 1$ . Then  $\alpha$  is irreducible if and only if  $d = 3$  and  $u \in \{-2 + b\sqrt{3} : b \in \{-1, 1\}\}$ .

*Proof.* Since  $d \equiv 2, 3 \pmod{4}$ , we have  $\mathcal{O}_K = \mathbb{Z}[\sqrt{d}]$ . Write  $u = a + b\sqrt{d}$ ; where  $a, b \in \mathbb{Z}$ . Then  $\alpha = (a + 1) + b\sqrt{d} \in \mathcal{O}_K = \mathbb{Z}[\sqrt{d}]$ . Consider  $N(\alpha)$ :

$$\begin{aligned}
N(\alpha) &= (a + 1)^2 - b^2d \\
&= a^2 + 2a + 1 - b^2d \\
&= (a^2 - b^2d) + 2a + 1 \\
&= N(u) + 2a + 1 \\
&= 1 + 2a + 1 \\
&= 2a + 2 \\
&= 2(a + 1)
\end{aligned}$$

Suppose that  $\alpha$  is prime. By Theorem 2.3.1,  $|N(\alpha)| = p$  or  $p^2$ .

**Case 1:** If  $|N(\alpha)| = p$  implies  $p = 2|a + 1|$  implies  $a = 0$  or  $a = -2$ .

If  $a = 0$ , then  $1 = N(u) = N(0 + b\sqrt{d}) = -b^2d$  implies  $b^2d = -1$ , only if  $d = -1$  and  $b = \pm 1$  which is a contradiction.

If  $a = -2$ , then  $1 = N(u) = N(-2 + b\sqrt{d}) = 4 - b^2d$  implies  $b^2d = 3$ , only if  $d = 3$  and  $b = \pm 1$ .

**Case 2:**  $|N(\alpha)| = p^2$  implies  $2|a + 1| = p^2$  implies  $a = 1$  or  $a = -3$ .  $\alpha$  need not be irreducible.

If  $a = -3$ , then using  $N(u) = 1$ , we get  $a^2 - db^2 = 1$ . Thus  $9 - db^2 = 1$ , so  $db^2 = 8$ . Hence  $d = 2$  and  $b = \pm 2$ .

Therefore  $\alpha = -2 \pm 2\sqrt{2} = 2(-1 \pm \sqrt{2})$ . But,  $2 = (2 - \sqrt{2})(2 + \sqrt{2})$ .

Thus,  $\alpha = \beta\gamma$  when  $\beta = 2 - \sqrt{2}$  and  $\gamma = (2 + \sqrt{2})(-1 \pm \sqrt{2})$ . Moreover,  $N(\beta) = 2$  and  $N(\gamma) = -2$ . Therefore, neither  $\beta$  nor  $\gamma$  is a unit, and hence  $\alpha$  is reducible.

On the other hand, if  $a = 1$ , then from  $N(u) = 1$ , we obtain  $1 - db^2 = 1$ , so  $b = 0$ . Thus  $u = 1$  and hence  $\alpha = u + 1 = 2$ . So,  $\alpha$  is reducible by Lemma 2.3.9.

Therefore  $d = 3$  and  $u \in \{-2 + b\sqrt{3} : b \in \{-1, 1\}\}$ .

Conversely, Let  $u \in \{-2 + b\sqrt{3} : b \in \{-1, 1\}\}$ . Then for each such  $u$ , we have  $N(u) = 1$ , since  $4 - 3b^2 = 4 - 3 = 1$ . Now, considering  $\alpha = u + 1 = -1 + b\sqrt{3}$  and hence  $|N(\alpha)| = |1 - 3b^2| = |1 - 3| = 2$  is a rational prime. Hence,  $\alpha$  is irreducible.  $\square$

**Example 2.3.7.** For  $\alpha = -1 + \sqrt{3}$ , we generated the following Mersenne primes in  $\mathbb{Q}(\sqrt{3})$  by using our own [Mathematica code](#):

| $p$ | $M_{p,\alpha}$                | $N(M_{p,\alpha})$ |
|-----|-------------------------------|-------------------|
| 2   | $\sqrt{3}$                    | -3                |
| 3   | $4 - \sqrt{3}$                | 13                |
| 7   | $154 - 87\sqrt{3}$            | 1009              |
| 13  | $63298 - 36543\sqrt{3}$       | 464257            |
| 19  | $26321410 - 15196671\sqrt{3}$ | 195941377         |

Table 2.3: Some Mersenne primes in  $\mathbb{Q}(\sqrt{3})$ .

The next three mersenne primes are found at  $p = 43$ , with

$$N(M_{p,\alpha}) = 5875117118252384257,$$

and at  $p = 61$  with

$$N(M_{p,\alpha}) = 422488668352620169103671297,$$

and at  $p = 151$  with

$$N(M_{p,\alpha}) = 812464719798480758896826989137871722017971967538570308893318053889.$$

and at  $p = 257$  with

$$N(M_{p,\alpha}) = 150526963475413836240783517481356521911317153399847$$

$$4678476114168570598\ 9693521659658490870320236091272657933172737.$$

**Remark 2.3.3.**

1. By Theorem 2.3.7, for the same prime integer  $p > 1$ , if the element  $\alpha$  produces a Mersenne prime, then its conjugate will also produce a Mersenne prime. Because of this, we do not need to test the conjugates.
2.  $2 - \sqrt{2}$ ,  $-\sqrt{2}$ , and  $-1 + \sqrt{3}$  are the conjugates of  $2 + \sqrt{2}$ ,  $\sqrt{2}$ , and  $-1 - \sqrt{3}$  respectively.
3. Theorem 2.3.8 and Theorem 2.3.10 imply the following: Among all fields  $\mathbb{Q}(\sqrt{d})$ ,  $d \equiv 2, 3 \pmod{4}$ ,  $\mathbb{Q}(\sqrt{2})$  and  $\mathbb{Q}(\sqrt{3})$  are the only fields where  $\alpha = 1 + u$  is irreducible. Therefore, it is sufficient to study Mersenne primes using only three choices for  $\alpha$ , namely  $\sqrt{2}$ ,  $2 + \sqrt{2}$  and  $-1 + \sqrt{3}$  by Theorem 2.3.7.

**Theorem 2.3.11.** Let  $d \equiv 1 \pmod{4}$  with  $d > 0$ . Let  $\alpha$  be an element in  $\mathcal{O}_K$  such that  $\alpha - 1 = u = \frac{a+b\sqrt{d}}{2}$  and  $N(\alpha - 1) = -1$ . If  $\alpha$  is irreducible, then one of the following statements is true:

1.  $|a|$  is an odd rational prime, and  $b$  is an odd integer.
2.  $|a|$  is square of an odd rational prime, and  $b$  is an odd integer.
3.  $|a| = 4$ , and in this case  $|b| = 2$  and  $d = 5$ .

*Proof.* Suppose that  $\alpha - 1 = u = \frac{a+b\sqrt{d}}{2}$  and  $N(u) = -1$ . Then:

$$N(u) = \frac{a^2 - db^2}{4} = -1$$

And so

$$a^2 - b^2d = -4$$

Since,  $\alpha = u + 1 = \frac{a+b\sqrt{d}}{2} + 1$ . We get

$$\alpha = \frac{a + b\sqrt{d}}{2} + 1 = \frac{a + 2 + b\sqrt{d}}{2}$$

Therefore,

$$N(\alpha) = \frac{(a+2)^2}{4} - \frac{b^2d}{4}$$

So,

$$N(\alpha) = \frac{a^2 - b^2d}{4} + a + 1$$

then we have

$$N(\alpha) = -1 + a + 1 = a$$

So, we get

$$|N(\alpha)| = |a|$$

Since  $\alpha$  is irreducible, its norm  $|N(\alpha)| = p$  or  $p^2$  where  $p$  is rational prime.

Therefore,  $|a| = p$  or  $p^2$  where  $p$  is rational prime.

By Lemma 2.2.19,  $a$  and  $b$  have the same parity. So we must consider two cases

If  $b$  is odd, then  $a$  is also odd, but  $|a| = p$  or  $p^2$  where  $p$  is rational prime, then  $|a|$  is odd rational prime or  $|a|$  is square of odd rational prime

If  $b$  is even, then  $a$  is even, then  $|a|$  is even, but  $|a| = p$  or  $p^2$  where  $p$  is rational prime, then  $|a| = 2$  or  $|a| = 4$

If  $|a| = 2$ , then  $a = 2$  or  $a = -2$ . In both cases,  $a^2 = 4$ .

using  $a^2 - b^2d = -4$  we get

$$4 - b^2d = -4$$

and hence

$$b^2d = 8$$

But  $d$  is a square free and  $d \equiv 1 \pmod{4}$ .

There is no integer solution for  $b^2d = 8$  with  $d \equiv 1 \pmod{4}$  since the positive divisors of 8 are 1, 2, 4, and 8, and we know that  $b^2 = 4$  or  $b^2 = 1$ , but  $b$  is even, so we get  $b^2$  is even, then  $b^2 = 4$ . Hence,  $d = 2$ . However,  $2 \equiv 2 \pmod{4}$ , which also contradicts  $d \equiv 1 \pmod{4}$ .

Since no positive divisors of 8 satisfy both  $d \equiv 1 \pmod{4}$  and the square-free condition, no such integer solution exists. Hence,  $|a| = 2$  cannot occur.

Hence,  $|a| = 4$ .

Therefore,  $a^2 = 16$ . Substitute this in the equation ( $a^2 - b^2d = -4$ ):

$$16 - b^2d = -4$$

So,

$$b^2d = 20$$

Since  $d \equiv 1 \pmod{4}$  and  $d > 0$ , the only solution for the equation  $b^2d = 20$  is  $b^2 = 4$ ,  $d = 5$  so,  $|b| = 2$ . □

Next, we discuss the converse of Theorem 2.3.11 and examine whether these conditions imply that  $\alpha$  is irreducible.

**Condition 1:** If  $|a|$  is an odd rational prime, then  $|N(\alpha)| = |a|$  is a rational prime, and hence  $\alpha$  is irreducible.

**Condition 2:** Let  $|a| = 4$ ,  $d = 5$ , and  $|b| = 2$ . Then  $\alpha - 1 = \frac{\pm 4 \pm 2\sqrt{5}}{2} = \pm 2 \pm \sqrt{5}$ . Thus,  $\alpha = (\pm 2 \pm \sqrt{5}) + 1$ . The values for  $\alpha$  are  $3 \pm \sqrt{5}$  and  $-1 \pm \sqrt{5}$ . Observe that  $N(3 \pm \sqrt{5}) = 9 - 5 = 4$  and  $N(-1 \pm \sqrt{5}) = 1 - 5 = -4$ . Assume for the sake of contradiction that  $\alpha$  is reducible, meaning  $\alpha = \beta\gamma$  where neither  $\beta$  nor  $\gamma$  is a unit in

$\mathcal{O}_K$ . Taking the norm gives  $N(\alpha) = \pm 4 = N(\beta)N(\gamma)$ . Since  $\gamma$  and  $\beta$  are not units, we must have  $N(\beta) = \pm 2$ . Let  $\beta = x_1 + y_1 \left( \frac{1+\sqrt{5}}{2} \right)$  with  $x_1, y_1 \in \mathbb{Z}$ . Then

$$N(\beta) = x_1^2 + x_1 y_1 - y_1^2 = \pm 2$$

By Lemma 2.2.19,  $x_1$  and  $y_1$  both are odd integers or  $x_1$  and  $y_1$  both are even integers.

If  $x_1$  and  $y_1$  both are odd integers, then  $x_1^2 + x_1 y_1 - y_1^2 \equiv 1 \pmod{2}$  so,  $\pm 2 \equiv 1 \pmod{2}$  and this is absurd. Hence, both  $x_1$  and  $y_1$  must be even integers. Let  $x_1 = 2k_1$  and  $y_1 = 2k_2$  for some  $k_1, k_2 \in \mathbb{Z}$ . Then

$$(2k_1)^2 + (2k_1)(2k_2) - (2k_2)^2 = 4k_1^2 + 4k_1 k_2 - 4k_2^2 = \pm 2$$

Dividing the entire equation by 2 gives:

$$2(k_1^2 + k_1 k_2 - k_2^2) = \pm 1$$

This implies an even integer equals an odd integer ( $\pm 1$ ), which is a contradiction. Thus, condition 2 implies  $\alpha$  is irreducible.

**Lemma 2.3.12.** Let  $d > 1$  be a square-free integer such that  $d \equiv 1 \pmod{4}$ , and let  $\mathcal{O}_K = \mathbb{Z} \left[ \frac{1+\sqrt{d}}{2} \right]$ . Let  $p$  be a rational prime, and  $\alpha \in \mathcal{O}_K$  satisfies  $N(\alpha) = \pm p^2$ . Then the following hold:

1. If there is no  $\beta \in \mathcal{O}_K$  such that  $N(\beta) = \pm p$ , then  $\alpha$  is irreducible in  $\mathcal{O}_K$ .
2. In particular, if the congruence  $x^2 + xy - \left( \frac{d-1}{4} \right) y^2 \equiv 0 \pmod{p}$  has only the trivial solution  $x \equiv y \equiv 0 \pmod{p}$ , then  $\alpha$  is irreducible in  $\mathcal{O}_K$ .

*Proof.*

1. Assume that  $\alpha$  is reducible. Then there exist non-unit  $\beta_1, \beta_2 \in \mathcal{O}_K$  such that  $\alpha = \beta_1 \beta_2$ . Hence  $N(\alpha) = N(\beta_1)N(\beta_2) = \pm p^2$ . Thus  $N(\beta_1) \neq \pm 1$  and  $N(\beta_2) \neq \pm 1$ . So  $N(\beta_1) = \pm p, N(\beta_2) = \pm p$ , and this is absurd, since there is no  $\beta$  in  $\mathcal{O}_K$  such that  $N(\beta) = \pm p$ .
2. Assume that  $x^2 + xy - \left( \frac{d-1}{4} \right) y^2 \equiv 0 \pmod{p}$  has only the trivial solution. Suppose  $\beta = x + y \left( \frac{1+\sqrt{d}}{2} \right) \in \mathcal{O}_K$  with  $N(\beta) = \pm p$ . Then  $x^2 + xy - \left( \frac{d-1}{4} \right) y^2 \equiv 0 \pmod{p}$ . So, the only solution of the congruence is  $x \equiv 0 \pmod{p}$  and  $y \equiv 0 \pmod{p}$ . Thus  $p \mid x$  and  $p \mid y$  and hence  $\beta = p \left( x_1 + y_1 \left( \frac{1+\sqrt{d}}{2} \right) \right)$ . Therefore  $p^2 \mid N(\beta)$ , which contradicts  $N(\beta) = \pm p$ . So, there is no  $\beta \in \mathcal{O}_K$  satisfying  $N(\beta) = \pm p$ . Thus  $\alpha$  is irreducible.

□

**Theorem 2.3.13.** Let  $d \equiv 1 \pmod{4}$ . Let  $u = \frac{a+b\sqrt{d}}{2}$  be a unit in  $\mathcal{O}_K$  such that  $N(u) = 1$ , and let  $\alpha = u + 1 = \frac{(a+2)+b\sqrt{d}}{2}$ . If  $\alpha$  is irreducible then one of the following conditions holds:

1.  $a + 2 = \pm p$  where  $p$  is an odd rational prime and  $b$  is an odd integer.
2.  $a + 2 = 4$  and  $b = 0$ .
3.  $a + 2 = p^2$  where  $p$  is an odd rational prime, and  $b^2d = p^2(p^2 - 4)$  with unique  $b$  and  $d$  in  $\mathbb{Z}^+$ .
4.  $a + 2 = -p^2$  where  $p$  is an odd rational prime, and  $b^2d = p^2(p^2 + 4)$  with unique  $b$  and  $d$  in  $\mathbb{Z}^+$ .

*Proof.* Let  $u = \frac{a+b\sqrt{d}}{2}$  and  $\alpha = u + 1 = \frac{a+2+b\sqrt{d}}{2}$  be an irreducible element. Then  $N(\alpha) = \frac{a^2+4a+4-b^2d}{4} = a + 2$ .

Since  $\mathcal{O}_K$  is (PID),  $N(\alpha) = \pm p$  or  $\pm p^2$  where  $p$  is a rational prime. and hence  $a + 2 = \pm p$  or  $a + 2 = \pm p^2$ .

**Case 1:**  $a + 2 = p$ .

If  $p = 2$ , then  $a = 0$ , which implies  $b^2d < 0$ , leading to a contradiction.

If  $p$  is an odd prime, then  $a + 2$  is odd, but since  $\alpha$  in  $\mathcal{O}_K$  and  $\alpha = \frac{(a+2)+b\sqrt{d}}{2}$ , by Lemma 2.2.19  $b$  should be an odd integer.

**Case 2:**  $a + 2 = -p$ .

If  $p = 2$ , then  $a = -4$ , which implies  $b^2d = 12$  and so  $d = 3$ , leading to a contradiction.

If  $p$  is an odd prime, then  $a + 2$  is odd, but since  $\alpha$  in  $\mathcal{O}_K$  and  $\alpha = \frac{(a+2)+b\sqrt{d}}{2}$ , by Lemma 2.2.19  $b$  should be an odd integer.

Thus,  $a + 2 = \pm p$  where  $p$  is an odd prime and  $b$  is an odd integer.

**Case 3:**  $a + 2 = p^2$ .

If  $p = 2$ , then  $a = 2$ . Since  $N(u) = 1$ , we get  $a^2 = b^2d + 4$ , we get  $b^2d = 0$ , but since  $d > 1$ ,  $b = 0$ , so we get  $\alpha = \frac{4+0\sqrt{d}}{2} = 2$ , Thus we get  $a + 2 = 4$  and  $b = 0$ .

If  $p$  is an odd prime, Using  $N(u) = 1$ , we get  $b^2d = a^2 - 4 = (p^2 - 2)^2 - 4 = p^2(p^2 - 4)$ . To show that the equation  $b^2d = p^2(p^2 - 4)$  has a unique solution in  $\mathbb{Z}$ .

We want to show  $p^2 - 4$  is not a perfect square. If  $p^2 - 4 = (p - 2)(p + 2)$  is a full square, then by using Theorem 1.1.5  $\gcd(p - 2, p + 2) = \gcd(p - 2, p - 2 + 4) = \gcd(p - 2, 4)$  and  $\gcd(p - 2, 4) = 1$  Since  $p - 2$  is odd integer, Now by using Theorem 1.1.18 both  $p - 2$  and  $p + 2$  would be full squares. Let  $p - 2 = k^2$  and  $p + 2 = m^2$  for odd integers  $k, m$ . Then  $4 = m^2 - k^2 = (m - k)(m + k)$ . The factors of 4 imply systems like  $m - k = \pm 2$  and  $m + k = \pm 2$ , and this implies  $m = \pm 2$ , which contradicts that  $m$  is odd. Thus,  $p^2 - 4$  is not a full square for an odd prime  $p$ , so  $d \neq 1$  by applying

Theorem 1.1.17 to the value  $n = p^2(p^2 - 4)$ , we can uniquely write  $n$  as a product of a square free integer  $d$  and a perfect square  $b^2$ .

Since  $p$  is an odd prime, we have  $p = 2k + 1$ , this implies

$$p^2 = (2k + 1)^2 = 4k^2 + 4k + 1 \equiv 1 \pmod{4}$$

Hence, it is clear that

$$p^2 - 4 \equiv 1 - 0 \equiv 1 \pmod{4}$$

Note that  $b^2d = p^2(p^2 - 4) \equiv 1 \pmod{4}$ . Since  $b$  is odd,  $b^2 \equiv 1 \pmod{4}$ , which implies

$$b^2d \equiv 1 \cdot d \equiv d \pmod{4}$$

Thus, we obtain

$$d \equiv 1 \pmod{4}$$

So, there exists a unique square free  $d > 1$  such that  $d \equiv 1 \pmod{4}$  and  $b$  such that  $b^2d = p^2(p^2 - 4)$ .

**Case 4:** Assume  $N(\alpha) = a + 2 = -p^2$ .

If  $p = 2$ , then  $a + 2 = -4$ , and hence  $a = -6$ . So Since  $N(u) = 1$ , we get  $a^2 = b^2d + 4$ , we get  $b^2d = 32$ , which implies  $d = 2$ , Contradicting  $d \equiv 1 \pmod{4}$ .

If  $p$  is an odd prime, we have

$$b^2d = a^2 - 4 = (-p^2 - 2)^2 - 4 = (p^2 + 2)^2 - 4 = p^2(p^2 + 4)$$

But  $p^2 + 4$  is not a perfect square for odd prime  $p$ , Since if  $p^2 + 4 = k^2$ , then  $k^2 - p^2 = 4$ ,  $(k - p)(k + p) = 4$ .

But  $p$  is odd, so  $k$  is also odd, and hence  $k - p, k + p$  are even. The only possible even factorization of is  $2 \cdot 2$  which give  $p = 0$ , a contradiction. Hence  $p^2 + 4$  is not square. Therefore the square free  $d > 1$  is unique determined.

□

This example illustrates why the condition  $N(\alpha) = p^2$  alone is not enough to decide whether  $\alpha$  is irreducible.

**Example 2.3.8.** This example illustrates how Lemma 2.3.12 can be used to test irreducibility when  $N(\alpha) = p^2$ . Assume that  $d \equiv 1 \pmod{4}$ . Let  $u = \frac{a+b\sqrt{d}}{2}$  be a unit in  $\mathcal{O}_K$  with  $N(u) = 1$ , and let  $\alpha = \frac{(a+2)\pm b\sqrt{d}}{2}$ . Then  $N(\alpha) = a + 2$ .

We consider several cases where  $N(\alpha) = p^2$ , with  $p$  a rational prime.

**Case 1:**  $p = 2$ . Let  $a + 2 = 4$ ,  $b = 0$  and  $d = 17$ . Then  $\alpha = 2 = \left(\frac{-3-\sqrt{17}}{2}\right) \left(\frac{3-\sqrt{17}}{2}\right)$  and  $N\left(\frac{\pm 3-\sqrt{17}}{2}\right) = -2$ . So  $\alpha$  is reducible. The congruence  $x^2 + xy - y^2 \left(\frac{17-1}{4}\right) \equiv 0$

(mod 2) has three solutions modulo 2.

$$\begin{array}{c|cc} x \backslash y & 0 & 1 \\ \hline 0 & \textcolor{red}{0} & \textcolor{red}{0} \\ 1 & 1 & \textcolor{red}{0} \end{array} \quad x^2 + xy - y^2 \left( \frac{d-1}{4} \right) \pmod{2}$$


---

**Case 2:**  $p = 3$ . Let  $a + 2 = 9$ ,  $b = 3$  and  $d = 5$ . Then  $\alpha = \frac{9 \pm 3\sqrt{5}}{2}$  and  $N(\alpha) = 9$ . The congruence  $x^2 + xy - y^2 \left( \frac{5-1}{4} \right) \equiv 0 \pmod{3}$  has only the zero solution, so  $\alpha$  is irreducible.

$$\begin{array}{c|ccc} x \backslash y & 0 & 1 & 2 \\ \hline 0 & \textcolor{red}{0} & 2 & 2 \\ 1 & 1 & 1 & 2 \\ 2 & 1 & 2 & 1 \end{array} \quad x^2 + xy - y^2 \left( \frac{d-1}{4} \right) \pmod{3}$$


---

**Case 3:**  $p = 5$ . Let  $a + 2 = 25$ ,  $b = 5$  and  $d = 21$ . Then  $\alpha = \frac{25 \pm 5\sqrt{21}}{2} = \left( \frac{1 \pm \sqrt{21}}{2} \right) \left( \frac{4 \pm \sqrt{21}}{2} \right)$  where  $N\left( \frac{1 \pm \sqrt{21}}{2} \right) = -5$  and  $N(4 \pm \sqrt{21}) = -5$ . Thus,  $\alpha$  is reducible.

Notice that the congruence  $x^2 + xy - y^2 \left( \frac{21-1}{4} \right) \equiv 0 \pmod{5}$  has nine solutions. Hence the sufficient condition in Lemma 2.3.12 is not satisfied.

$$\begin{array}{c|ccccc} x \backslash y & 0 & 1 & 2 & 3 & 4 \\ \hline 0 & \textcolor{red}{0} & \textcolor{red}{0} & \textcolor{red}{0} & \textcolor{red}{0} & \textcolor{red}{0} \\ 1 & 1 & 2 & 3 & 4 & \textcolor{red}{0} \\ 2 & 4 & 1 & 3 & \textcolor{red}{0} & 2 \\ 3 & 4 & 2 & \textcolor{red}{0} & 3 & 1 \\ 4 & 1 & \textcolor{red}{0} & 4 & 3 & 2 \end{array} \quad x^2 + xy - y^2 \left( \frac{d-1}{4} \right) \pmod{5}$$


---

**Case 4:**  $p = 7$ . Let  $a + 2 = 49$ ,  $b = 21$  and  $d = 5$ . Then  $\alpha = \frac{49 \pm 21\sqrt{5}}{2}$  and  $N(\alpha) = 49$ . Moreover, the congruence  $x^2 + xy - y^2 \equiv 0 \pmod{7}$  has only the zero solution, so  $\alpha$  is irreducible.

| $x \backslash y$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|------------------|---|---|---|---|---|---|---|
| 0                | 0 | 6 | 3 | 5 | 5 | 3 | 6 |
| 1                | 1 | 1 | 6 | 2 | 3 | 2 | 6 |
| 2                | 4 | 5 | 4 | 1 | 3 | 3 | 1 |
| 3                | 2 | 4 | 4 | 2 | 5 | 6 | 5 |
| 4                | 2 | 5 | 6 | 5 | 2 | 4 | 4 |
| 5                | 4 | 1 | 3 | 3 | 1 | 4 | 5 |
| 6                | 1 | 6 | 2 | 3 | 2 | 6 | 1 |

$$x^2 + xy - y^2 \left( \frac{d-1}{4} \right) \pmod{7}$$


---

**Remark 2.3.4.**

1. In Example 2.3.8, we saw that when  $N(\alpha) = p^2$ , the element  $\alpha$  may be reducible or irreducible. Lemma 2.3.12 gives a useful sufficient condition for irreducibility.
2. Theorem 2.3.13 gives necessary conditions for the irreducibility of  $\alpha$ . condition (1) is also sufficient, because in this case  $N(\alpha) = a+2 = \pm p$  is a rational prime and hence  $\alpha$  is irreducible. However, in conditions (2), (3) and (4), the norm of  $\alpha$  is equal to  $\pm p^2$ .

**Example 2.3.9.** Let

$$\alpha = \frac{(a+2) \pm b\sqrt{d}}{2} \in \mathbb{Z} \left[ \frac{1 + \sqrt{d}}{2} \right]$$

The following table lists some cases with  $p < 100$  for which  $N(\alpha) = p^2$  by using [Mathematica](#).

In each case, the congruence

$$x^2 + xy - y^2 \left( \frac{d-1}{4} \right) \equiv 0 \pmod{p}$$

has only the trivial solution. Hence, they are irreducible.

| Prime ( $p$ ) | $a+2$ | $b$ | $d$ | Number of Zeros |
|---------------|-------|-----|-----|-----------------|
| 3             | 9     | 3   | 5   | 1               |
| 7             | 49    | 21  | 5   | 1               |
| 11            | 121   | 33  | 13  | 1               |
| 23            | 529   | 115 | 21  | 1               |
| 47            | 2209  | 987 | 5   | 1               |
| 79            | 6241  | 711 | 77  | 1               |

Table 2.4

**Example 2.3.10.** Here we search the primality of  $N(M_{q,\alpha})$  for various values of  $q < 100$  by using [Mathematica](#). The results focus on cases where the norm is a rational prime. The general form of  $\alpha$  that used is:

$$\alpha = \frac{(a+2) + b\sqrt{d}}{2}$$

The following table records some cases in which  $N(M_{q,\alpha})$  is prime

| Element                       | Prime | $N(M_{q,\alpha})$ (Prime Integer)   |
|-------------------------------|-------|-------------------------------------|
| $\frac{9+3\sqrt{5}}{2}$       | 2     | 19                                  |
| $\frac{9+3\sqrt{5}}{2}$       | 11    | 24,365,506,969                      |
| $\frac{49+21\sqrt{5}}{2}$     | 79    | Rational prime with 134 digits      |
| $\frac{121+33\sqrt{13}}{2}$   | 5     | 1062936601                          |
| $\frac{2209+987\sqrt{5}}{2}$  | 11    | 30378291542031571572830856662086369 |
| $\frac{2209+987\sqrt{5}}{2}$  | 29    | Rational prime with 96 digits       |
| $\frac{6241+711\sqrt{77}}{2}$ | 17    | Rational prime with 62 digits       |
| $\frac{6241+711\sqrt{77}}{2}$ | 59    | Rational prime with 222 digits      |

Table 2.5: Some Mersenne primes in real quadratic fields with  $d \equiv 1 \pmod{4}$ .

**Example 2.3.11.** This example shows that if  $q > 100$ , then we can find a Mersenne prime  $M_{q,\alpha}$  where  $\alpha = \frac{9+3\sqrt{5}}{2}$  and prime  $q = 373$  or  $389$ , by using our [Mathematica code](#).

## 2.4 Mathematica Codes

In this section, we provide the Mathematica code used to generate the results presented in Examples 2.3.5, 2.3.7, 2.3.9, 2.3.10 and 2.3.11.

Consider the real quadratic field  $\mathbb{Q}(\sqrt{2})$  and  $\alpha = 2 + \sqrt{2}$ . The following Mathematica code will show that the smallest rational prime  $p$  such that  $p > 233$  and  $N(M_{p,\alpha})$  is rational prime is  $p = 4057$ , and will compute the Mersenne prime analogue and its norm for  $p = 4057$  and the given  $\alpha$ . Note that 4057 is the 560th prime integer, which is evaluated by the code `Prime[560]`.

In[1]:=

```
(* 1. Define the base element and the initial functions *)
α= 2 + Sqrt[2];
j[x_] := x /. Sqrt[2] -> -Sqrt[2];
k[x_] := Expand[x * j[x]];
Mer[p_, x_] := FullSimplify[ExpandAll[(x^p - 1)/(x - 1)]];

(* 2. Searching for the prime that satisfies the Mersenne prime analogue
definition.*)
For[p = NextPrime[233], PrimeQ[k[Mer[p, α]]] != True, p=NextPrime[p]];
Print["Mersenne prime found for p = ", p];

(* 3. Compute the Mersenne prime analogue and its norm for the prime p and α. *)
Print["Mersenne Prime Analogue (M_p) = "];
Print[Mer[p, α]];
Print["Norm of M_p = ", k[Mer[p, α]]];
Print["PrimeQ of the Norm = ", PrimeQ[k[Mer[p, α]]]];
```

Out[1]:=

Mersenne prime found for (p) = 4057

Mersenne Prime Analogue ( $M_p$ ) =

```
752481694000090296985215101476573896110387654970239271870134684450846198674
351179656550142725695076231800027415121615658679188836652377113723891584657
494239734747621685664820982719444094411315687171721146610371672978053198916
013649203417970441091994572459073712508561820070109251331996720009229545324
750080644802976548605900631076020817377819412498004749178683143493396550036
351628624299878601368942807966820719981077320805327029733536878621906177360
210735655572284067355459697276759772802748102842063417088958865574619268994
532878368618261384993927861393688754300544341901151413266601033707004334661
753229172671039453152145183619336942789006667582447292147791038050565386528
131736621896394089804347850650552258183013873982996362569857394542994856777
1801424802852731851438545592531761182123326405008256015812295439879830426
207966847419171361003077336602453943190587851929142103609062245921678716610
922183526387657808555265733015868749257096975946546399776048061042832912019
475397796182982981085679989048973080987241950597804783908275353417221143378
590866670899571472197237689147143674430113544899903994683969484080050733496
775436311666067873843235956349420952693206216115982580915035351068050419769
781405737883833089577340828563659751594456909677748535797070183443232746203
```

906314353775006680372558603861107736595716298333768546268044325145081950451  
 323484523571921507681030098325736732092999241686272838978274914335841448938  
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 537410093933071339108143178586598270479397017006788830794053248399998353440  
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 199606668650882219262544609235245716130355875483123553846010629760854172890  
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 748255151882351880366522284217903151568943065328156156543113795622046975175  
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 464079388753961280207239288501813288794825742063270386372444161 +  
 532084908546204460783220000303747462107872672930764619130534242737210830316  
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 604492703623688987728285016194788935826789421444671756375850450165860958828  
 899308600352647829790628489713367625692556494813450791912368435018010717832  
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 654309172595572938838969754725447089290547344453881473881036221737348923765  
 944194772684303594502343591292033231588945237201751638990706092922552243169  
 637486439374219548056316844336886840543477334177480411319378636202674774800  
 789735138967450880103197710266341454142787665348002981342865307762442477723  
 840374050253121972551080793606121008017022223656437538361682779886005931540  
 55736868863832211622577027767705303862340217463414298140810426142618207331  
 579347095571684897147451274704137339129433439227388638460027947742539816779  
 296533645260054909085488260824668615148001040166922462508182421695999176106  
 50007374799200627207644947330985020173882852069773569111486809405410563434  
 840045034453645177333901114863162803370029376217184404662807698048095416708  
 6934311552886246785443660022464811084478256898714731964466866499435488043158  
 458151619511354943699014804704620836457130369428780776064603027572182031481  
 080321751217454380513612448362802222909525050066705269593720818304320439871  
 72979080272834575606854011724913699981640140327801330467443708430659065218  
 515604679312701219979443559414147713070724251043767966461448298596194904321  
 902852544849483302674411143586219674630113188532615207870812494821133606616  
 165333035152885216744291454603005789275077913511044305840330559319880284288  
 208561569708741159874596426053166379264714079420475930574614247568465976594  
 140927812270818681839992194420577371935086017627511458116212468915254316714  
 673873500815083170823897920356141889104330035824445899952572296123187769029  
 958470964864872305474383268182189483482056342523553444953849855 $\sqrt{2}$

Norm of  $M_p =$

3633303022184998437103310204168137640652266001663537020262406339850535718  
 613779501501578815747337492843026192215767571214825077725795419835943990120  
 839018916908497366685024688220032958357726400474690215975104955636575403397  
 627752810090478976652038064178969293156980516321236491872736032406251324589  
 691812642623678342043016960970777220948913791216234337124253624975755231351  
 451121228063094513153651025680712797183269312327389341084970475557907256190  
 036018705712555159453873433353406990048375722184937930001827391471221281409  
 3196904378574278628597579789857733427229013974048874454617138259269578062

```

634123695035724370661660169386140462242740202712980420479819726963050526468
120040252334530002557618070066469237442721808696772999281102539160776198686
232910230718067424562512976495853899598598616989519938157309139135690458760
055574874250708146242418945514270273702952326365250296132589585722893128424
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685863088762797668909468615802427038029470212792679133455648907058742902994
452967722040775749612827622196340369103676264724822535307893230859192649177
749557936471040403769333842522582506495101673453473997994072683534221422368
817788455883281763301909289324561848179999508356475062567600943378931748386
544761202124149128386603737782376062373270464163922354315466914564941079163
446643099878475632516873261860467487143202588701782962277966994194326116672
72457177718639304399013432558603801463171914860636522187490375068657232141
338229653894030567482621364852135374136434254325096263605282213768260446617
70480738582648281552662677096411466249882829949517548153641649740105521540
38417860590077784300625891480770516170796037200855987468440662948450733287
063872785850775817339777711868705359375257265487770095251065915202864000763
68621260315982188487069961930176553898122315777095046273224280479791156553
011983417199615346214672865978853828171083720082927352739402359798092120011
086541608116075842810817896452981765403632471666354836641446071044663987313
783222276608012548364212997658565431228258353941513566495983895470855049727
92880977142963392999681907737269513955912293871402285373295951871

```

PrimeQ of the Norm = True

In[1]:=

```

dPID = {2, 3, 5, 6, 7, 11, 13, 14, 17, 19, 21, 22, 23, 29, 31, 33, 37, 38, 41,
        43, 46, 47, 53, 57, 59, 61, 62, 67, 69, 71, 73, 77, 83, 86, 89, 93, 94, 97};

```

The following Mathematica code computes all prime integers  $p < 100$  and their corresponding integers  $d$  such that  $d \equiv 1 \pmod{4}$ , where  $d$  satisfies the equation  $b^2d = p^2(p^2 - 4)$  for some integer  $b$ .

In[2]:=

```

B = Table[{Prime[i], Apply[Times, Map[#[[1]]^Mod[#[[2]], 2] &,
    FactorInteger[(Prime[i]^2 - 4) Prime[i]^2]]]}, {i, 2, PrimePi[100]}]

```

Out[2]=

```

{{3, 5}, {5, 21}, {7, 5}, {11, 13}, {13, 165}, {17, 285}, {19, 357}, {23, 21}, {29, 93},
{31, 957}, {37, 1365}, {41, 1677}, {43, 205}, {47, 5}, {53, 2805}, {59, 3477}, {61,
413}, {67, 4485}, {71, 5037}, {73, 213}, {79, 77}, {83, 85}, {89, 7917}, {97, 1045}}

```

The following code generates a list for odd primes  $p < 100$ , containing  $p$ ,  $p^2$ , the positive integer  $b$ , and the integer  $d$  (where  $d \equiv 1 \pmod{4}$ ), such that  $b^2d = p^2(p^2 - 4)$ , along with the number of solutions  $(x, y)$  to the equation  $x^2 + xy - y^2 \frac{(d-1)}{4} \equiv 0 \pmod{p}$ .

In[3]:=

```

Cc = Table[{
  Prime[i + 1],
  Prime[i + 1]^2,
  Sqrt[(Prime[i + 1]^2 - 4) Prime[i + 1]^2 / Apply[Times, Map#[[1]]^Mod#[[2]],
    2] &, FactorInteger[(Prime[i + 1]^2 - 4)]]],
  Apply[Times, Map#[[1]]^Mod#[[2]], 2] &, FactorInteger[(Prime[i + 1]^2 -
    4)]]],
  Length[Solve[(x + y/2)^2 - B[[i, 2]] y^2 / 4 == 0, {x, y}, Modulus -> Prime[i
    + 1]]]
}, {i, 1, Length[B]};

```

Out[3]=

```

{{3, 9, 3, 5, 1}, {5, 25, 5, 21, 9}, {7, 49, 21, 5, 1}, {11, 121, 33, 13, 1}, {13, 169,
13, 165, 25}, {17, 289, 17, 285, 33}, {19, 361, 19, 357, 1}, {23, 529, 115, 21, 1}, {29,
841, 87, 93, 57}, {31, 961, 31, 957, 1}, {37, 1369, 37, 1365, 73}, {41, 1681, 41, 1677,
81}, {43, 1849, 129, 205, 1}, {47, 2209, 987, 5, 1}, {53, 2809, 53, 2805, 105}, {59,
3481, 59, 3477, 1}, {61, 3721, 183, 413, 121}, {67, 4489, 67, 4485, 1}, {71, 5041, 71,
5037, 1}, {73, 5329, 365, 213, 145}, {79, 6241, 711, 77, 1}, {83, 6889, 747, 85, 1},
{89, 7921, 89, 7917, 177}, {97, 9409, 291, 1045, 193}}

```

The following code filters the list *Cc* to select elements where the value of *d* belongs to the set dPID.

In[4]:=

```

A = Select[Cc, Union[dPID, #[[4]]] === dPID&]

```

Out[4]=

```

{{3, 9, 3, 5, 1}, {5, 25, 5, 21, 9}, {7, 49, 21, 5, 1}, {11, 121, 33, 13, 1}, {23, 529,
115, 21, 1}, {29, 841, 87, 93, 57}, {47, 2209, 987, 5, 1}, {79, 6241, 711, 77, 1}}

```

The list *A1* is then generated by mapping the selected elements to the form  $\alpha = \frac{p^2 + b\sqrt{d}}{2}$ , where *p*, *b*, and *d* are extracted from the list *A*. This represents the elements  $\alpha$  in the quadratic field  $\mathbb{Q}(\sqrt{d})$  derived from the set *A* such that the solution  $(x, y)$  to the equation  $x^2 + xy - y^2 \frac{(d-1)}{4} \equiv 0 \pmod{p}$  is only the trivial solution.

In[5]:=

```

(* For Example 2.3.9*)
A1 = Map[(#[[2]] + #[[3]] * Sqrt#[[4]]) / 2 &, Select[A, #[[5]] == 1 &]]

```

Out[5]=

```

{1/2(9 + 3√5), 1/2(49 + 21√5), 1/2(121 + 33√13), 1/2(529 + 115√21), 1/2(2209 + 987√5), 1/2(6241 + 711√77)}

```

In[6]:=

```

(* Here are two ways to write the Mersenne prime analogue in real quadratic
   fields using Mathematica *)

f[α_, p_] := (α^p - 1) / (α - 1) // ExpandAll // FullSimplify

(* Or we can use *)

f1[α_, p_] := Module[{β1 = Expand[α^p]-1, β2=α-1}, (β1/β2) // FullSimplify]

(* Here are two ways to write the norm function in Mathematica: *)

normoff1[α_] :=
  If[Head[Expand[α]] === Plus,
    Expand[α][[1]]^2 - Expand[α][[2]]^2,
    If[Head[α] === Integer ||
      Head[α] === Rational, α^2, -α^2]]

(* Or we can use *)

normoff2[α_] :=
  Module[{β=Expand[2α]}, If[Head[β] === Plus, β[[1]]^2 - β[[2]]^2,
    If[Head[β] === Integer, β^2, -β^2]]]/4

g[α_] := Module[{set = {}, β= α},
  For[i = 1, Prime[i] <= Prime[100], i++,
    If[PrimeQ[normoff1[f[α, Prime[i]]]] == True,
      AppendTo[set, {β, Prime[i], Length[IntegerDigits[normoff1[f[α,
        Prime[i]]]]], normoff1[f[α, Prime[i]]]}]
    ]
  ];
  set
]

(* For Example 2.3.7*)
Grid[
  Join[
    {"α", "p", "Length of N(Mp,α)", "N(Mp,α)"}, g[-1+Sqrt[3]] ],
  Frame -> True,
  Dividers -> All,
  Alignment -> {Center, Baseline},
  Background -> Lighter[Gray, 0.5]
]

```

Out[6]=

| $\alpha$        | $p$ | Length of $N(M_{p,\alpha})$ | $N(M_{p,\alpha})$                                                                                                              |
|-----------------|-----|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| $-1 + \sqrt{3}$ | 2   | 1                           | -3                                                                                                                             |
| $-1 + \sqrt{3}$ | 3   | 2                           | 13                                                                                                                             |
| $-1 + \sqrt{3}$ | 7   | 4                           | 1009                                                                                                                           |
| $-1 + \sqrt{3}$ | 13  | 6                           | 464257                                                                                                                         |
| $-1 + \sqrt{3}$ | 19  | 9                           | 195941377                                                                                                                      |
| $-1 + \sqrt{3}$ | 43  | 19                          | 5875117118252384257                                                                                                            |
| $-1 + \sqrt{3}$ | 61  | 27                          | 422488668352620169103671297                                                                                                    |
| $-1 + \sqrt{3}$ | 151 | 66                          | 81246471979848075889682698913<br>78717220179719675385703088933<br>18053889                                                     |
| $-1 + \sqrt{3}$ | 257 | 113                         | 15052696347541383624078351748<br>13565219113171533998474678476<br>114168570598 9693521659658490<br>870320236091272657933172737 |

In[7]:=

```
(* And we can find the Mersenne prime analogues by the following code *)
Map[f[-1+Sqrt[3],#]&,{2,3,7,13,19,43,61,151,257}];
```

In[8]:=

```
(* And for Example 2.3.5*)
Grid[
  Join[
    {" $\alpha$ ", "p", "Length of  $N(M_{p,\alpha})$ ", " $N(M_{p,\alpha})$ "}, g[2 + Sqrt[2]]],
  Frame -> True,
  Dividers -> All,
  Alignment -> {Center, Baseline},
  Background -> Lighter[Gray, 0.5]
]
```

Out[8]=

| $\alpha$       | $p$ | Length of $N(M_{p,\alpha})$ | $N(M_{p,\alpha})$                                                                                                                             |
|----------------|-----|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| $2 + \sqrt{2}$ | 2   | 1                           | 7                                                                                                                                             |
| $2 + \sqrt{2}$ | 3   | 2                           | 31                                                                                                                                            |
| $2 + \sqrt{2}$ | 5   | 3                           | 431                                                                                                                                           |
| $2 + \sqrt{2}$ | 7   | 4                           | 5279                                                                                                                                          |
| $2 + \sqrt{2}$ | 11  | 6                           | 732799                                                                                                                                        |
| $2 + \sqrt{2}$ | 73  | 39                          | 85156905517225879321860274148<br>0913108991                                                                                                   |
| $2 + \sqrt{2}$ | 89  | 48                          | 29031588678119168146433038877<br>2329064268797313023                                                                                          |
| $2 + \sqrt{2}$ | 233 | 125                         | 18060475427282023033368001231<br>16644178473780689153780654706<br>53141679119595184985817477128<br>29157156517940837234519177963<br>497324543 |

In[9]:=

```
(* And we can find the Mersenne prime analogues by the following code *)
Map[f[2+Sqrt[2],#]&,{2,3,5,7,11,73,89,233}];
```

This code searches for  $\alpha$  from the list A1 and rational prime  $p < 100$  that produce Mersenne primes. For each result, it displays the value of the norm and its length (number of digits):

In[10]:=

```
merseenepime = Table[Select[g[A1[[i]]], #[[2]] < 100 &], {i, 1, Length[A1]};
Length[merseenepime]
```

Out[10]=

6

In[11]:=

```
(* For Example 2.3.10*)
Grid[
  Join[
    {" $\alpha$ ", "q", "Length of  $N(M_{q,\alpha})$ ", " $N(M_{q,\alpha})$ "},
    merseenepime[[1]], merseenepime[[2]], merseenepime[[3]],
    merseenepime[[4]], merseenepime[[5]],merseenepime[[6]]
  ],
  Frame -> True,
  Dividers -> All,
```

```

Alignment -> {Center, Baseline},
Background -> Lighter[Gray, 0.5]
]

```

Out[11]:=

| $\alpha$                           | $q$ | Length of $N(M_{q,\alpha})$ | $N(M_{q,\alpha})$                                                                                                                                                                                                                                          |
|------------------------------------|-----|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\frac{1}{2}(9 + 3\sqrt{5})$       | 2   | 2                           | 19                                                                                                                                                                                                                                                         |
| $\frac{1}{2}(9 + 3\sqrt{5})$       | 11  | 11                          | 24365506969                                                                                                                                                                                                                                                |
| $\frac{1}{2}(49 + 21\sqrt{5})$     | 79  | 134                         | 27183421660571866808506712460<br>99310491600040466587435339833<br>33403556288849145502467753495<br>16615600167562859273939753775<br>448446269755054849                                                                                                     |
| $\frac{1}{2}(121 + 33\sqrt{13})$   | 5   | 10                          | 1062936601                                                                                                                                                                                                                                                 |
| $\frac{1}{2}(2209 + 987\sqrt{5})$  | 11  | 35                          | 3037829154203157157283085666208636                                                                                                                                                                                                                         |
| $\frac{1}{2}(2209 + 987\sqrt{5})$  | 29  | 96                          | 12511909498757453885707050229<br>26029566859886350922496970477<br>75966020392169519148055331996<br>698654241                                                                                                                                               |
| $\frac{1}{2}(6241 + 711\sqrt{77})$ | 17  | 62                          | 89955712581961944424004781747<br>03836694961119268046631502927<br>7281                                                                                                                                                                                     |
| $\frac{1}{2}(6241 + 711\sqrt{77})$ | 59  | 222                         | 78279367044023448146975719187<br>87732861926440858644630036557<br>81680330538011278059466451111<br>91489208854460576265525011851<br>69031050754377195037895958126<br>44920543230091442052521519953<br>56545874371699897966329789092<br>3557444582584201761 |

In[12]:=

```
A1[[1]]
```

Out[12]:=

$\frac{1}{2}(9 + 3\sqrt{5})$   
In[13]:=

*(\* For Example 2.3.11\*)*

`g[A1[[1]]][[-1]]`

`g[A1[[1]]][[-2]]`

**Out[13]:=**

$\{\frac{1}{2}(9+3\sqrt{5}), 389, 372, 1586120476795813409616411764216342963939984624038795$   
090420394707730678 991980472248788484844566147407424326367725339647275687  
8349334362830103 13423608481044342785278581222673961281013682334199737408  
20822305060951 3066525593808431616106580598457536037152165534921578580402  
506105032861 927787057119674596265177561150238612815662634426714316066553  
3544225152 4944535747536392216041}

$\{\frac{1}{2}(9+3\sqrt{5}), 373, 356, 8559650274391221606472429850579732089042022068684861$   
067121023049015499 735993544646894158288186196108738012205802656121400720  
8966164404589903 11032883265196560940737666690915759371292194384392188658  
74741441284083 8119717598715095965921357767748743078456876332703298527822  
866946559582 522773665718661457120842286858856917122403093797238204994539  
7109659700 704641}

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